JavaScript String Reference

‹ Previous

Next ›

SHARE

See Also:

Full JavaScript Strings Tutorial

Complete String Reference

Revised July 2025

Name	Description
at()	Returns an indexed character from a string
charAt()	Returns the character at a specified index (position)
charCodeAt()	Returns the Unicode of the character at a specified index
codePointAt()	Returns the Unicode value at an index (position) in a string
concat()	Returns two or more joined strings
constructor	Returns the string's constructor function
endsWith()	Returns if a string ends with a specified value
fromCharCode()	Returns Unicode values as characters
includes()	Returns if a string contains a specified value

indexOf()	Returns the index (position) of the first occurrence of a value in a string
isWellFormed()	Returns true if a string is well formed
lastIndexOf()	Returns the index (position) of the last occurrence of a value in a string
length	Returns the length of a string
localeCompare()	Compares two strings in the current locale
match()	Searches a string for a value, or a regular expression, and returns the matches
matchAll()	Searches a string for a value, or a regular expression, and returns the matches
padEnd()	Pads a string at the end
padStart()	Pads a string from the start
prototype	Allows you to add properties and methods to an object
repeat()	Returns a new string with a number of copies of a string
replace()	Searches a string for a pattern, and returns a string where the first match is replaced
replaceAll()	Searches a string for a pattern and returns a new string where all matches are replaced
search()	Searches a string for a value, or regular expression, and returns the index (position) of the match
slice()	Extracts a part of a string and returns a new string
split()	Splits a string into an array of substrings
startsWith()	Checks whether a string begins with specified characters
substr()	Deprecated. Use substring() or slice() instead.
substring()	Extracts characters from a string, between two specified indices (positions)
toLocaleLowerCase()	Returns a string converted to lowercase letters, using the host's locale
toLocaleUpperCase()	Returns a string converted to uppercase letters, using the host's locale
toLowerCase()	Returns a string converted to lowercase letters
toString()	Returns a string or a string object as a string
toUpperCase()	Returns a string converted to uppercase letters
toWellFormed()	Returns a string where "lone surrogates" are replaced with the Unicode replacement character
trim()	Returns a string with removed whitespaces
trimEnd()	Returns a string with removed whitespaces from the end
trimStart()	Returns a string with removed whitespaces from the start
valueOf()	Returns the primitive value of a string or a string object

Note

All string methods return a new value.

They do not change the original string.

REMOVE ADS

String HTML Wrapper Methods

HTML wrapper methods return a string wrapped inside an HTML tag.

These are not standard methods, and may not work as expected.

The HTML wrapper methods are deprecated in JavaScript.

They are only standardized for compatibility purposes and are not recommended for new development.

Modern web development practices advocate for using CSS for styling and DOM manipulation to create and modify HTML elements, providing greater control and flexibility.

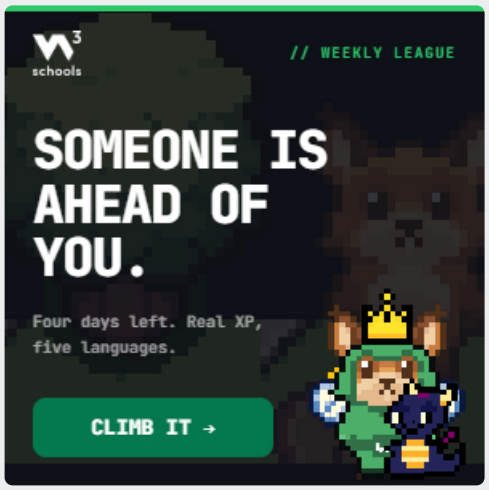
Method	Description
anchor()	Displays a string as an anchor
big()	Displays a string using a big font
blink()	Displays a blinking string
bold()	Displays a string in bold
fixed()	Displays a string using a fixed-pitch font
fontcolor()	Displays a string using a specified color
fontsize()	Displays a string using a specified size
italics()	Displays a string in italic
link()	Displays a string as a hyperlink
small()	Displays a string using a small font

strike()	Displays a string with a strikethrough
sub()	Displays a string as subscript text
sup()	Displays a string as superscript text

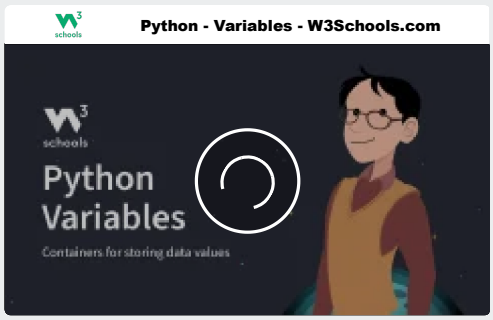
‹ Previous

Sign in to track progress

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

Spaces 0
League —
Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript while Loop

‹ Previous JavaScript Statements Next ›

Example

Loop a code block as long as a i is less than 5:

```
let text = "";
let i = 0;
while (i < 5) {
  text += i + "<br>";
  i++;
}
```

Try it Yourself »

Loop (iterate over) an array to collect car names:

```
const cars = ["BMW", "Volvo", "Saab", "Ford"];
let text = "";
let i = 0;
while (i < cars.length) {
  text += cars[i] + "<br>";
  i++;
}
```

Try it Yourself »

The loop starts in position 0 (`let i = 0`).

The loop increments `i` for each run (`i++`).

The loop runs as long as `i < cars.length`.

More examples below.

Description

SHARE

The `while` statement creates a loop (araund a code block) that is executed while a condition is `true`.

The loop runs while the condition is `true`. Otherwise it stops.

See Also:

The JavaScript while Tutorial

JavaScript Loop Statements

Statement	Description
break	Breaks out of a loop
continue	Skips a value in a loop
while	Loops a code block while a condition is true
do...while	Loops a code block once, and then while a condition is true
for	Loops a code block while a condition is true
for...of	Loops the values of any iterable
for...in	Loops the properties of an object

Syntax

```
while (condition) {
  code block to be executed
}
```

Parameters

Parameter	Description

condition	Required. The condition for running the code block. If it returns true, the code clock will start over again, otherwise it ends.

Note

If the condition is always true, the loop will never end. This will crash your browser.

If you use a variable in the condition, you must initialize it before the loop, and increment it within the loop. Otherwise the loop will never end. This will also crash your browser.

REMOVE ADS

More Examples

Loop over an array in descending order (negative increment):

```
const cars = ["BMW", "Volvo", "Saab", "Ford"];
let text = "";
let len = cars.length;
while (len--) {
  text += cars[len] + "<br>";
}
```

Try it Yourself »

Using break - Loop through a block of code, but exit the loop when i == 3:

```
let text = "";
let i = 0;
while (i < 5) {
  text += i + "<br>";
  i++;
  if (i == 3) break;
}
```

Try it Yourself »

Using continue - Loop through a block of code, but skip the value 3:

```
let text = "";
let i = 0;
while (i < 5) {
  i++;
  if (i == 3) continue;
  text += i + "<br>";
}
```

Try it Yourself »

Browser Support

`while` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

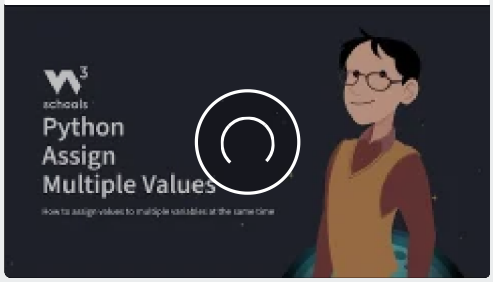
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Statements Next ›



REMOVE ADS

Python - Assign Multiple Values to Vari...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

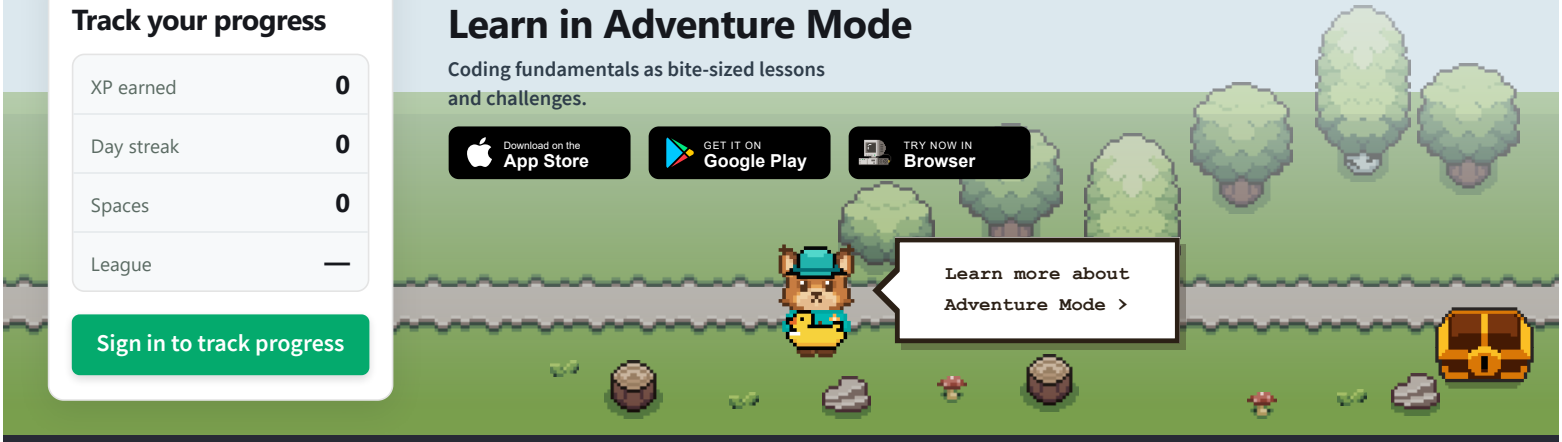
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String at()

‹ Previous JavaScript String Reference Next ›

Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at(0);
```

Try it Yourself »

Get the first character of the text:

```
let text = "W3Schools";
let character = text[0];
```

Try it Yourself »

More examples below.

Description

SHARE

The `at()` method returns an indexed character from a string.

The `at()` method returns the same as `[]`.

Syntax

```
string.at(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) of the character to be returned. Default is 0. -1 returns the last character.

Return Value

Type	Description
A String	A single character. The character in the given position (index) in the string.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

JavaScript String `at()` is supported in all browsers since March 2022:

				
Chrome 66	Edge 79	Firefox 61	Safari 12	Opera 50
Jul 2021	Jul 2021	Jul 2021	Mar 2022	Aug 2021

REMOVE ADS

More Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at();
```

Try it Yourself »

Get the last character of the text:

```
let text = "W3Schools";
let character = text.at(-1);
```

Try it Yourself »

‹ Previous

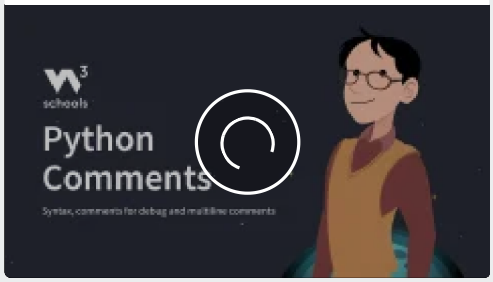
JavaScript String Reference

Next ›



REMOVE ADS

Python - Comments - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

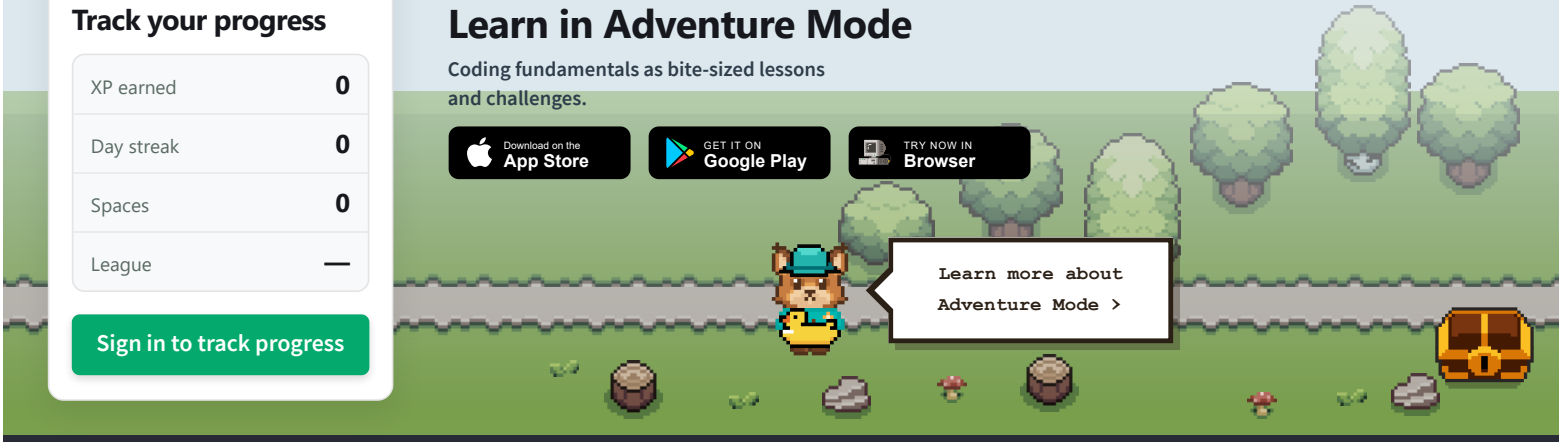
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String at()

‹ Previous JavaScript String Reference Next ›

Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at(0);
```

Try it Yourself »

Get the first character of the text:

```
let text = "W3Schools";
let character = text[0];
```

Try it Yourself »

More examples below.

Description

SHARE

The `at()` method returns an indexed character from a string.

The `at()` method returns the same as `[]`.

Syntax

```
string.at(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) of the character to be returned. Default is 0. -1 returns the last character.

Return Value

Type	Description
A String	A single character. The character in the given position (index) in the string.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

JavaScript String `at()` is supported in all browsers since March 2022:

				
Chrome 66	Edge 79	Firefox 61	Safari 12	Opera 50
Jul 2021	Jul 2021	Jul 2021	Mar 2022	Aug 2021

REMOVE ADS

More Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at();
```

Try it Yourself »

Get the last character of the text:

```
let text = "W3Schools";
let character = text.at(-1);
```

Try it Yourself »

‹ Previous

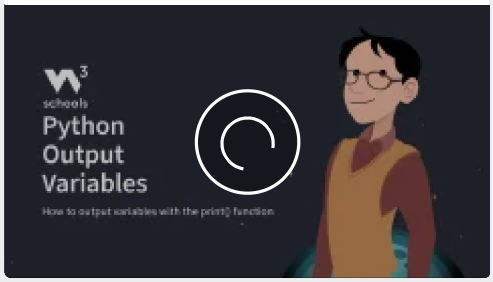
JavaScript String Reference

Next ›



REMOVE ADS

Python - Output Variables with print() - ...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

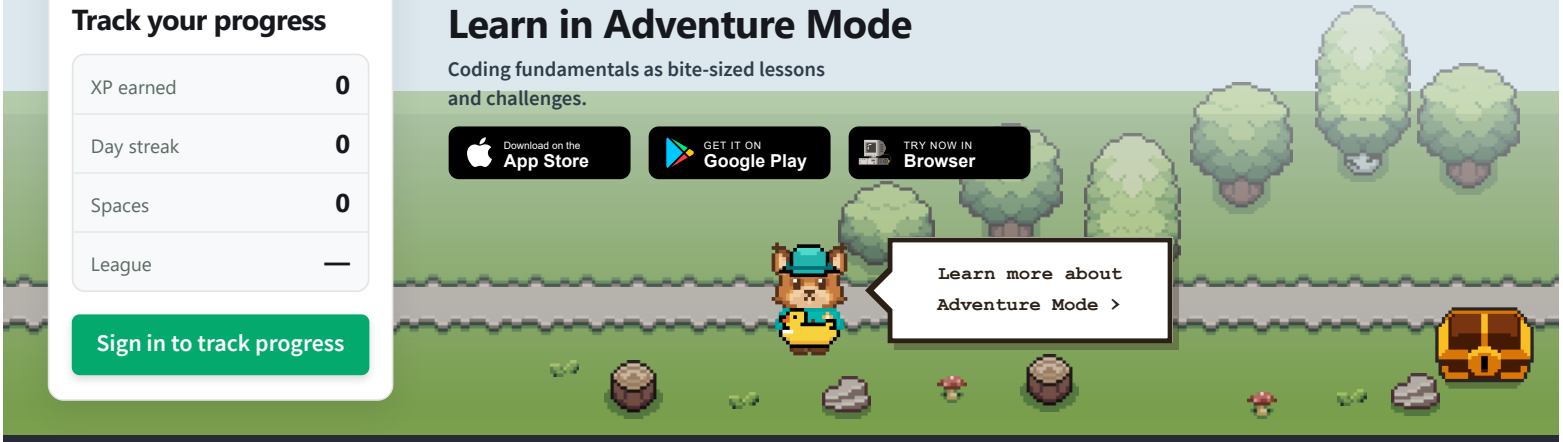
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String charAt()

‹ Previous JavaScript String Reference Next ›

Examples

Get the **first** character in a string:

```
let text = "HELLO WORLD";
let letter = text.charAt(0);
```

Try it Yourself »

Get the **second** character in a string:

```
let text = "HELLO WORLD";
let letter = text.charAt(1);
```

Try it Yourself »

Get the **last** character in a string:

```
let text = "HELLO WORLD";
let letter = text.charAt(text.length-1);
```

Try it Yourself »

More examples below.

Description

SHARE

The `charAt()` method returns the character at a specified index (position) in a string.

The index of the first character is 0, the second 1, ...

See Also:

The charCodeAt() Method

The codePointAt() Method

The indexOf() Method

The lastIndexOf() Method

Syntax

```
string.charAt(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) of the character. Default = 0.

Return Value

Type	Description
String	The character at the specified index. Empty string ("") if the index is out of range.

REMOVE ADS

More Examples

Index out of range returns empty string:

```
let text = "HELLO WORLD";
let letter = text.charAt(15);
```

Try it Yourself »

Default index is 0:

```
let text = "HELLO WORLD";
let letter = text.charAt();
```

Try it Yourself »

Invalid index converts to 0:

```
let text = "HELLO WORLD";
let letter = text.charAt(3.14);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`charAt()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

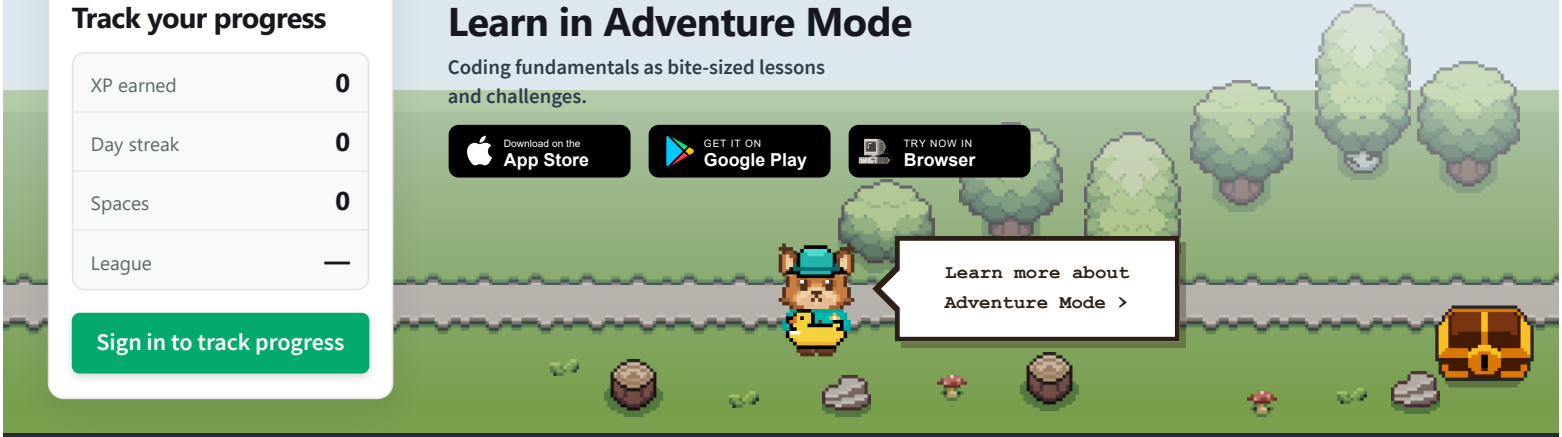
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String charCodeAt()

‹ Previous | JavaScript String Reference | Next ›

Examples

Get the Unicode of the first character in a string:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(0);
```

Try it Yourself »

Get the Unicode of the second:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(1);
```

Try it Yourself »

More examples below.

Description

SHARE

The `charCodeAt()` method returns the Unicode of the character at a specified index (position) in a string.

The index of the first character is 0, the second is 1, ...

The index of the last character is string length - 1 (See Examples below).

See also the `charAt()` method.

See Also:

The charAt() Method

The codePointAt() Method

The indexOf() Method

The lastIndexOf() Method

charCodeAt() vs codePointAt()

`charCodeAt()` is UTF-16, `codePointAt()` is Unicode.

`charCodeAt()` returns a number between 0 and 65535.

Both methods return an integer representing the UTF-16 code of a character, but only `codePointAt()` can return the full value of a Unicode value greather 0xFFFF (65535).

For more information about Unicode Character Sets, visit our Unicode Reference.

Syntax

```
string.charCodeAt(index)
```

Parameters

Parameter	Description
index	Optional. A number. The index (position) of a character. Default value = 0.

Return Value

Type	Description
A number	The Unicode of the character at the specified index. NaN if the index is invalid.

REMOVE ADS

More Examples

Get the Unicode of the last character in a string:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(text.length-1);
```

Try it Yourself »

Get the Unicode of the 15th character:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(15);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`charCodeAt()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

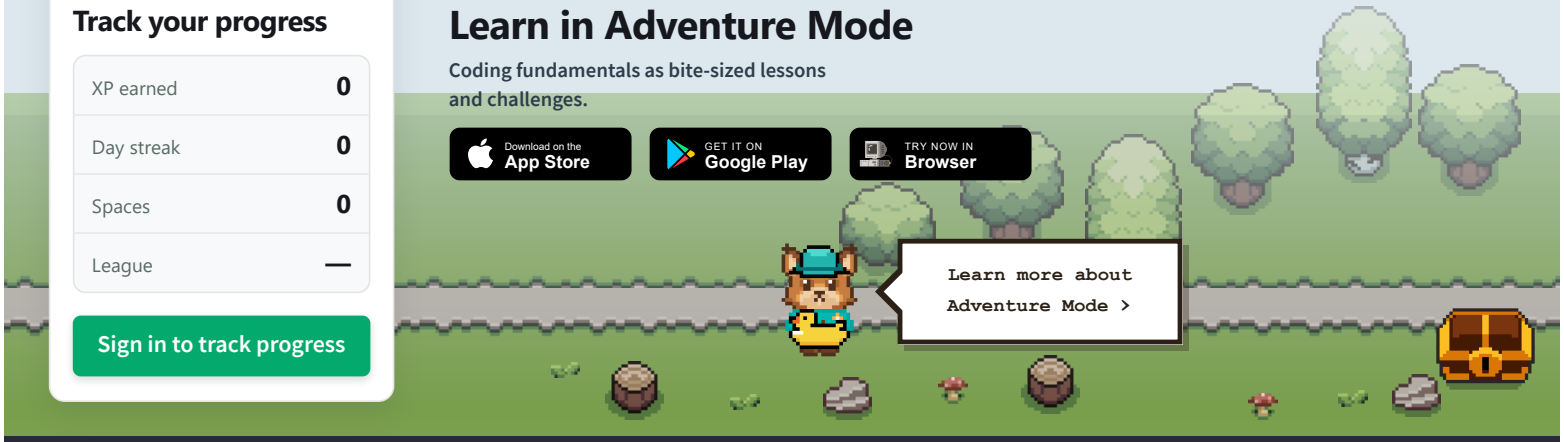
NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser

Learn more about Adventure Mode >



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String codePointAt()

‹ Previous JavaScript String Reference Next ›

Examples

Get code point value at the first position in a string:

```
let text = "HELLO WORLD";
let code = text.codePointAt(0);
```

Try it Yourself »

Get the code point value at the second position:

```
let text = "HELLO WORLD";
let code = text.codePointAt(1);
```

Try it Yourself »

More examples below.

Description

SHARE

The `codePointAt()` method returns the Unicode value at an index (position) in a string.

The index of the first position is 0, the second is 1, ...

See Also:

The charCodeAt() Method

The charAt() Method

The indexOf() Method

The lastIndexOf() Method

Unicode

For more information about Unicode Character Sets, visit our Unicode Reference.

Difference Between charCodeAt() and codePointAt()

`charCodeAt()` is UTF-16, `codePointAt()` is Unicode.

`charCodeAt()` returns a number between 0 and 65535.

Both methods return an integer representing the UTF-16 code of a character, but only `codePointAt()` can return the full value of a Unicode value greather 0xFFFF (65535).

For more information about Unicode Character Sets, visit our Unicode Reference.

Syntax

```
string.codePointAt(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) in a the string. Default value = 0.

Return Value

Type	Description

Number	The code point value at the specified index. `undefined` if the index is invalid.

REMOVE ADS

More Examples

Get the code point value at the last position:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(text.length-1);
```

Try it Yourself »

Get the code point value at the 15th position:

```
let text = "HELLO WORLD";
let code = text.charCodeAt(15);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`codePointAt()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38

May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous

JavaScript String Reference

Next ›

W3 schools

JS

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

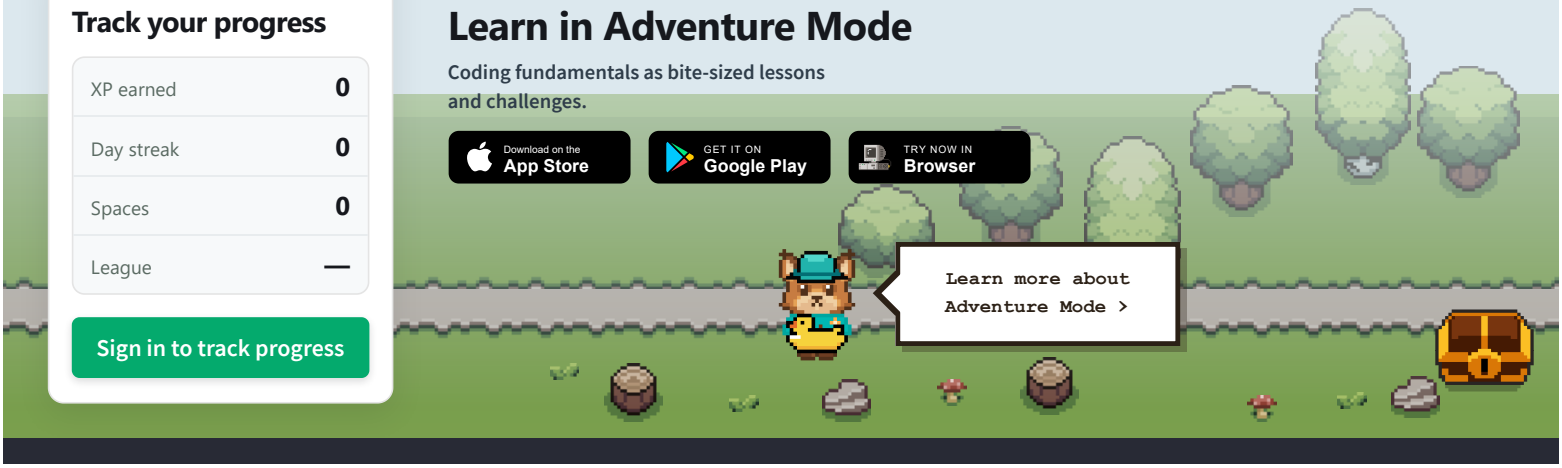
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String concat()

‹ Previous JavaScript String Reference Next ›

Examples

Join two strings:

```
let text1 = "sea";
let text2 = "food";
let result = text1.concat(text2);
```

Try it Yourself »

Join two strings:

```
let text1 = "Hello";
let text2 = "world!";
let result = text1.concat(" ", text2);
```

Try it Yourself »

More examples below.

Description

SHARE

The `concat()` method joins two or more strings.

The `concat()` method does not change the existing strings.

The `concat()` method returns a new string.

Syntax

```
string.concat(string1, string2, ..., stringX)
```

Parameters

Parameter	Description
string1, *string2,* ... *stringX*	Required. The strings to be joined.

Return Value

Type	Description
A string	A new string containing the combined strings.

REMOVE ADS

More Examples

Join three strings:

```
let text1 = "Hello";
let text2 = "world!";
let text3 = "Have a nice day!";
let result = text1.concat(" ", text2, " ", text3);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`concat()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›



REMOVE ADS



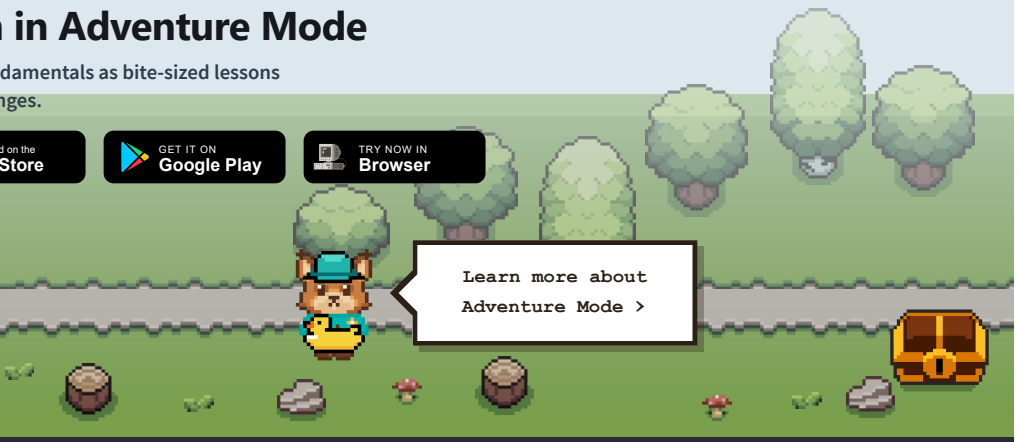
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String constructor

‹ Previous JavaScript String Reference Next ›

Example

Get the constructor of a string:

```
let message = "Hello World!";
let text = message.constructor;
```

Try it Yourself »

Description

SHARE

The `constructor` property returns the function that created the String prototype.

For JavaScript strings the `constructor` returns:

function String() { [native code] }

Syntax

```
string.constructor
```

Return Value

```
function String() { [native code] }
```

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`constructor` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

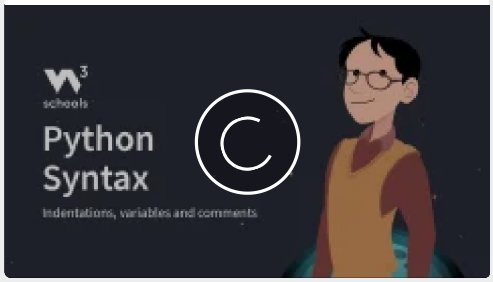
JavaScript String Reference

Next ›



REMOVE ADS

Python - Syntax - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

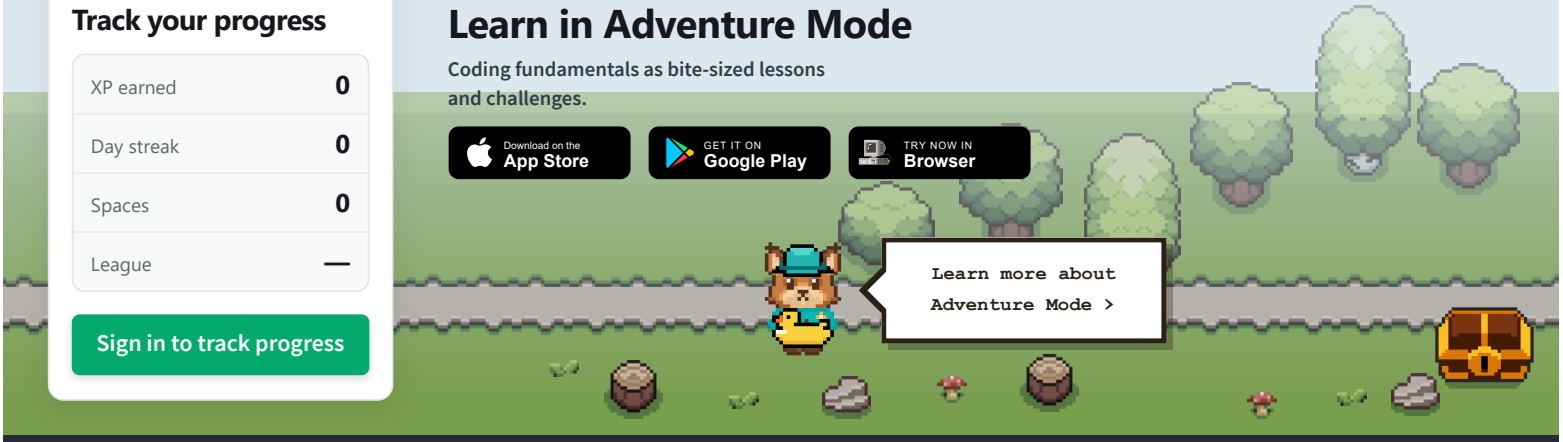
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String endsWith()

‹ Previous JavaScript String Reference Next ›

Examples

Check if a string ends with "world":

```
let text = "Hello world";
let result = text.endsWith("world");
```

Try it Yourself »

```
let text = "Hello World";
let result = text.endsWith("world");
```

Try it Yourself »

More examples below.

Description

SHARE

The `endsWith()` method returns `true` if a string ends with a specified string.

Otherwise it returns `false`.

The `endsWith()` method is case sensitive.

See Also:

The startsWith() Method

Syntax

```
string.endsWith(searchvalue, length)
```

Parameters

Parameter	Description
searchvalue	Required. The string to search for.
length	Optional. The length of the string to search. Default value is the length of the string.

Return Value

Type	Description
A boolean	`true` if the string ends with the value, otherwise `false`.

REMOVE ADS

More Examples

Check if the 11 first characters of a string ends with "world":

```
let text = "Hello world, welcome to the universe.";
text.endsWith("world", 11);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`endsWith()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript String Reference | Next ›

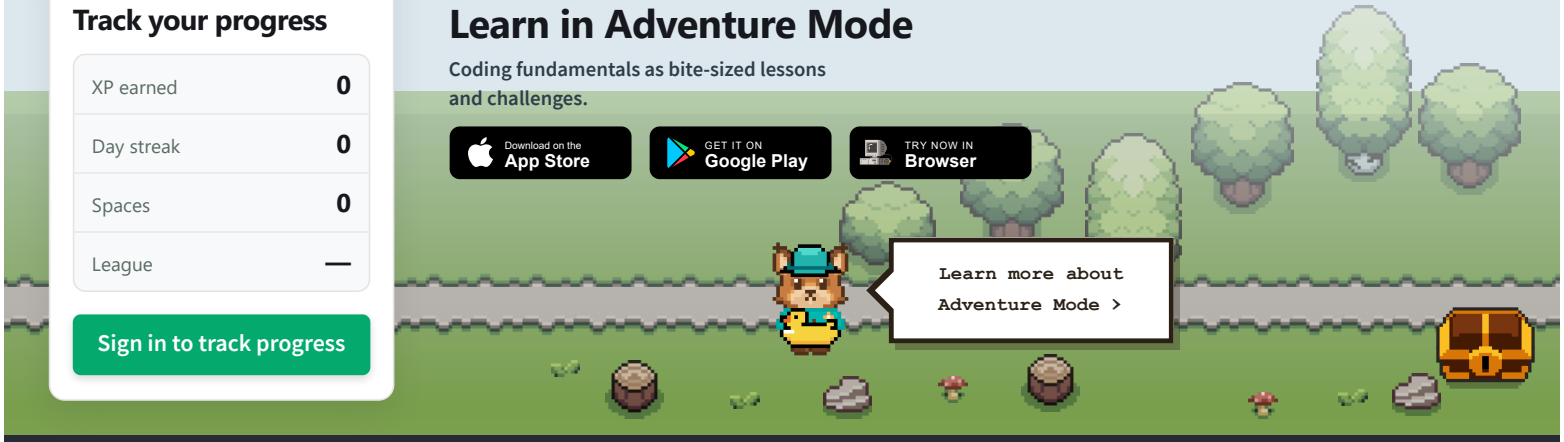


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



PLUS
SPACES
GET CERTIFIED
FOR TEACHERS
PRACTICE
CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String.fromCharCode()

‹ Previous JavaScript String Reference Next ›

Examples

How to convert Unicode values to characters:

```
let char = String.fromCharCode(65);
```

Try it Yourself »

```
let text = String.fromCharCode(72, 69, 76, 76, 79);
```

Try it Yourself »

Description

SHARE

The `String.fromCharCode()` method converts Unicode values to characters.

The `String.fromCharCode()` is a static method of the String object.

The syntax is always `String.fromCharCode()`.

You cannot use `myString.fromCharCode()`.

Syntax

```
String.fromCharCode(n1, n2, ..., nX)
```

Parameters

Parameters	Description
n1, *n2*, *nX*	Required. One or more Unicode values to be converted.

Return Value

Type	Description
A string	A string representing the unicode character(s).

REMOVE ADS

Tip

For a list of all Unicode values, please study our Complete Unicode Reference.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`String.fromCharCode()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:



Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript String Reference | Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



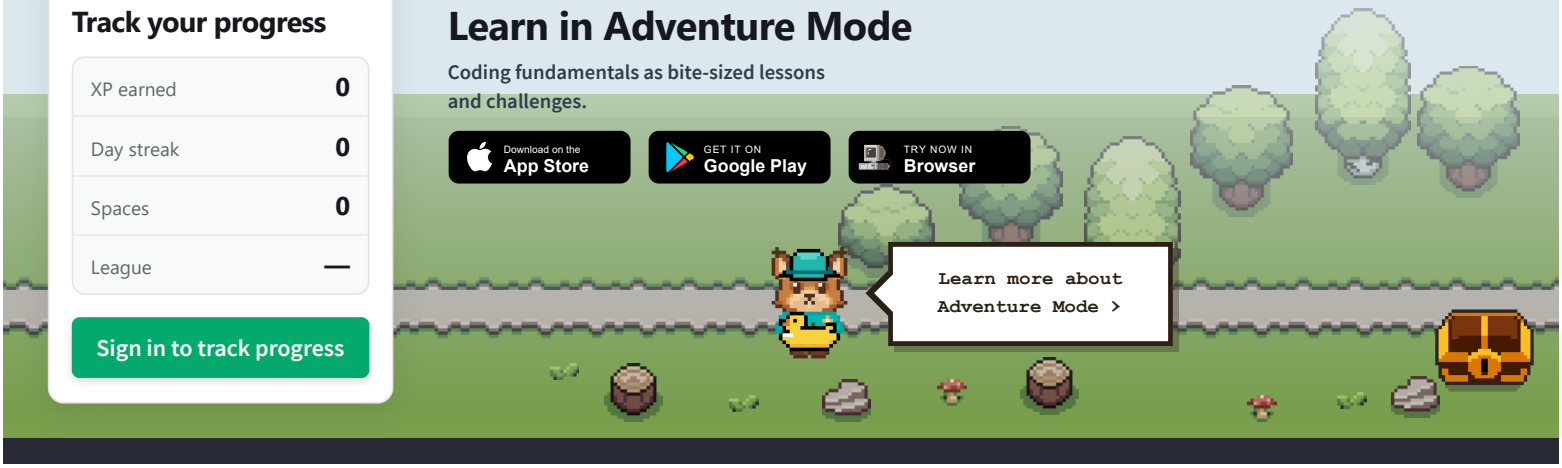
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String includes()

‹ Previous JavaScript String Reference Next ›

Examples

Check if a string includes "world":

```
let text = "Hello world, welcome to the universe.";
let result = text.includes("world");
```

Try it Yourself »

More examples below.

Description

SHARE

The `includes()` method returns `true` if a string contains a specified string.

Otherwise it returns `false`.

The `includes()` method is case sensitive.

Syntax

```
string.includes(searchvalue, start)
```

Parameters

Parameter	Description
searchvalue	Required. The string to search for.
start	Optional. The position to start from. Default value is 0.

Return Value

Type	Description
A boolean.	`true` if the string contains the value, otherwise `false`.

REMOVE ADS

More Examples

Start at position 12:

```
let text = "Hello world, welcome to the universe.";
let result = text.includes("world", 12);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`includes()` is an ECMAScript6 (ES6 2015) feature.

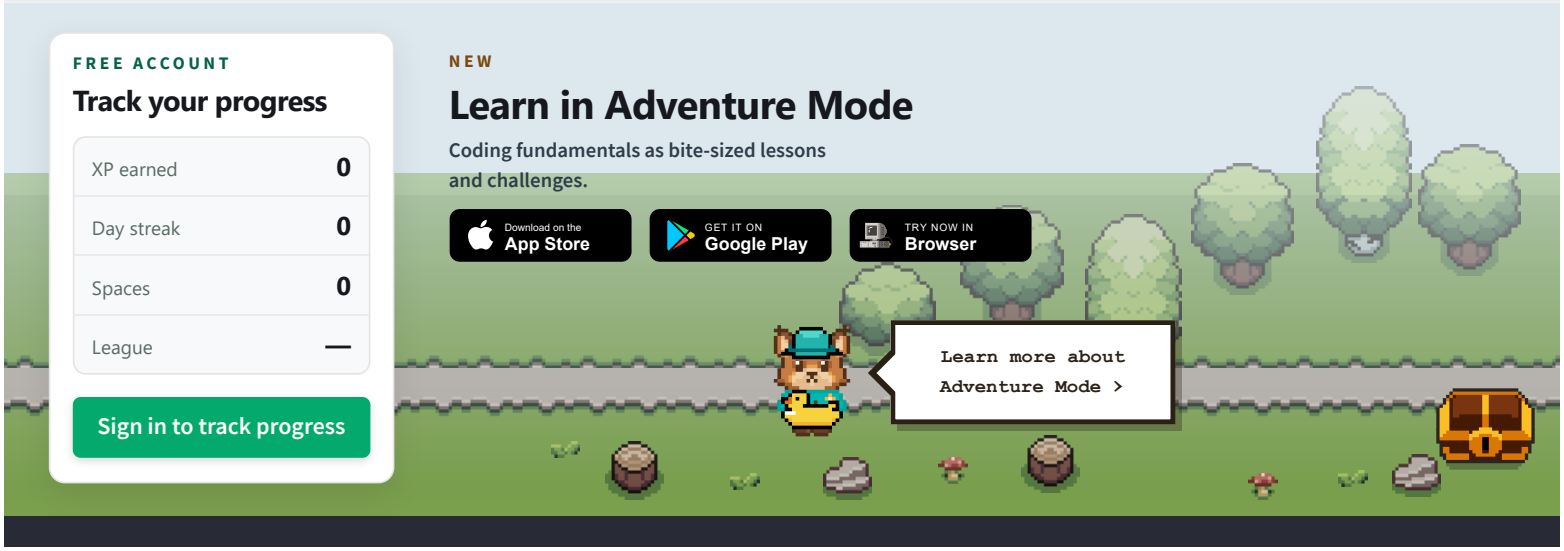
JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript String Reference | Next ›



REMOVE ADS



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String indexOf()

‹ Previous JavaScript String Reference Next ›

Examples

Search a string for "welcome":

```
let text = "Hello world, welcome to the universe.";
let result = text.indexOf("welcome");
```

Try it Yourself »

Search a string for "Welcome":

```
let text = "Hello world, welcome to the universe.";
let result = text.indexOf("Welcome");
```

Try it Yourself »

Find the first occurrence of "e":

```
let text = "Hello world, welcome to the universe.";
text.indexOf("e");
```

Try it Yourself »

Find the first occurrence of "e", starting at position 5:

```
let text = "Hello world, welcome to the universe.";
text.indexOf("e", 5);
```

Try it Yourself »

Find the first occurrence of "a":

```
let text = "Hello world, welcome to the universe.";
text.indexOf("a");
```

Try it Yourself »

Description

SHARE

The `indexOf()` method returns the position of the first occurrence of a value in a string.

The `indexOf()` method returns -1 if the value is not found.

The `indexOf()` method is case sensitive.

See Also:

The lastIndexOf() Method

The search() Method

The match() Method

Syntax

```
string.indexOf(searchvalue, start)
```

Parameters

Parameter	Description
searchvalue	Required. The string to search for.
start	Optional. The position to start from (default is 0).

Return Value

Type	Description
A number	The first position where the search-value occurs. -1 if it never occurs.

REMOVE ADS

The Difference Between String indexOf() and String search()

The `indexOf()` method cannot search against a regular expression.

The `search()` cannot take a start position argument.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`indexOf()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

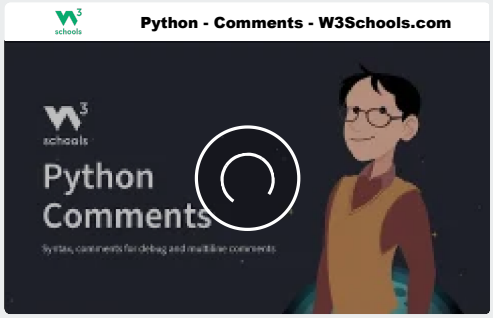
Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

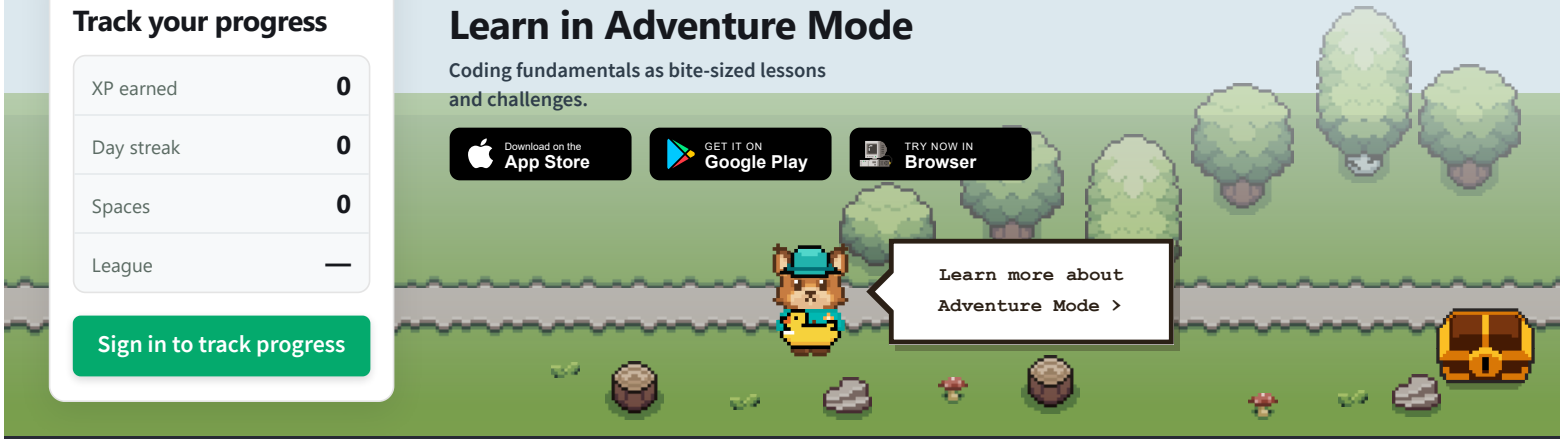
NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser

Learn more about Adventure Mode >



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String isWellFormed()

‹ Previous JavaScript String Reference Next ›

Examples

```
let text = "Hello world!";
let result = text.isWellFormed();
```

Try it Yourself »

```
let text = "Hello World \uD800";
let result = text.isWellFormed();
```

Try it Yourself »

Description

SHARE

The `isWellFormed()` method returns `true` if a string is well formed.

Otherwise it returns `false`.

A string is well formed if does not contain **lone surrogates**.

Lone Surrogates

A **Lone surrogate** is a Unicode surrogate code that is not a part of a valid surrogate pair, used to represent characters in UTF-16 encoding.

Syntax

```
string.isWellFormed()
```

Parameters

NONE

Return Value

Type	Description
Boolean	`true` if the string is well formed, otherwise `false`.

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`isWellFormed()` is a JavaScrip 2024 feature.

ES 2024 is supported in all modern browsers since **March 2024**:

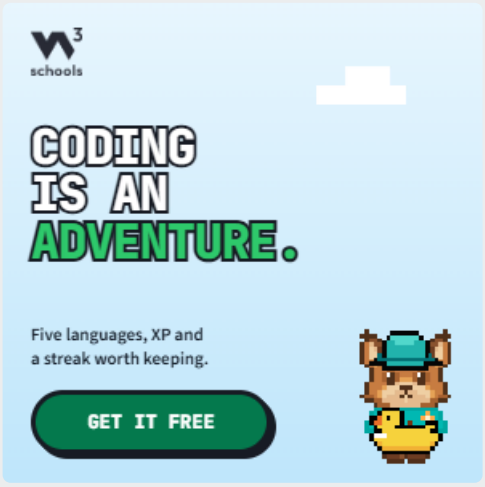
				
Chrome 117	Edge 117	Firefox 119	Safari 17.4	Opera 103

Sep 2023	Sep 2023	Oct 2023	Okt 2024	May 2023

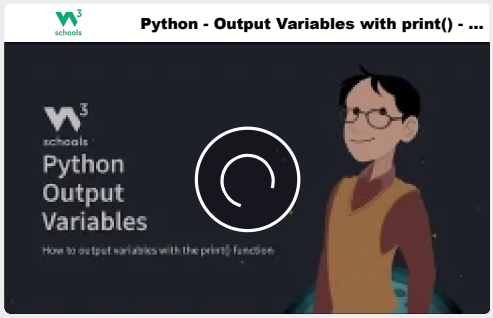
‹ Previous

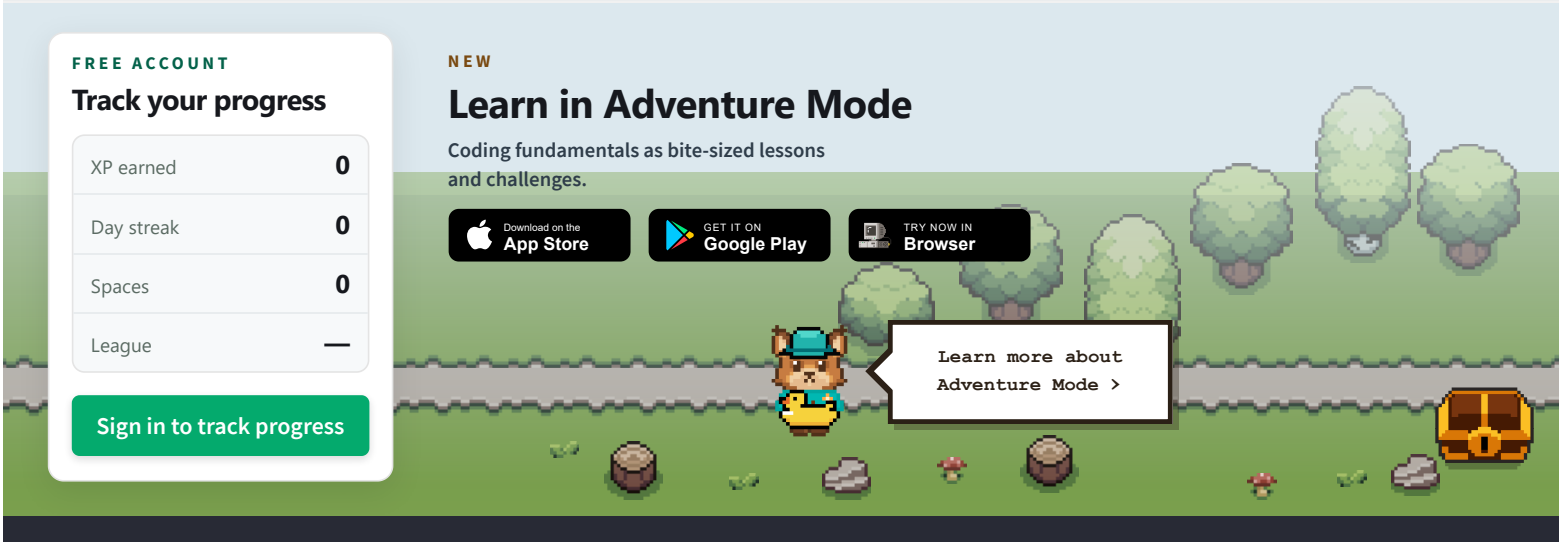
JavaScript String Reference

Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String lastIndexOf()

‹ Previous JavaScript String Reference Next ›

Examples

Search for the last occurrence of "planet":

```
let text = "Hello planet earth, you are a great planet.";
let result = text.lastIndexOf("planet");
```

Try it Yourself »

```
let text = "Hello planet earth, you are a great planet.";
let result = text.lastIndexOf("Planet");
```

Try it Yourself »

More examples below.

Description

SHARE

The `lastIndexOf()` method returns the index (position) of the last occurrence of a specified value in a string.

The `lastIndexOf()` method searches the string from the end to the beginning.

The `lastIndexOf()` method returns the index from the beginning (position 0).

The `lastIndexOf()` method returns -1 if the value is not found.

The `lastIndexOf()` method is case sensitive.

See Also:

The indexOf() Method

Syntax

```
string.lastIndexOf(searchvalue, start)
```

Parameters

Parameter	Description
searchvalue	Required. The string to search for.
start	Optional. The position where to start. Default value is string length.

Return Value

Type	Description
A number	The position where the search-value occurs. -1 if it never occurs.

REMOVE ADS

More Examples

Search for the last occurrence of "planet", starting at position 20:

```
let text = "Hello planet earth, you are a great planet.";
let result = text.lastIndexOf("planet", 20);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`lastIndexOf()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›

Learn it.
Get certified.
Get hired.

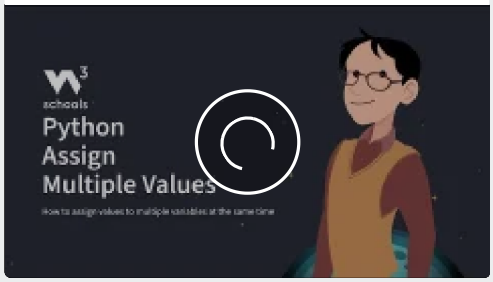
$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS

Python - Assign Multiple Values to Vari...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

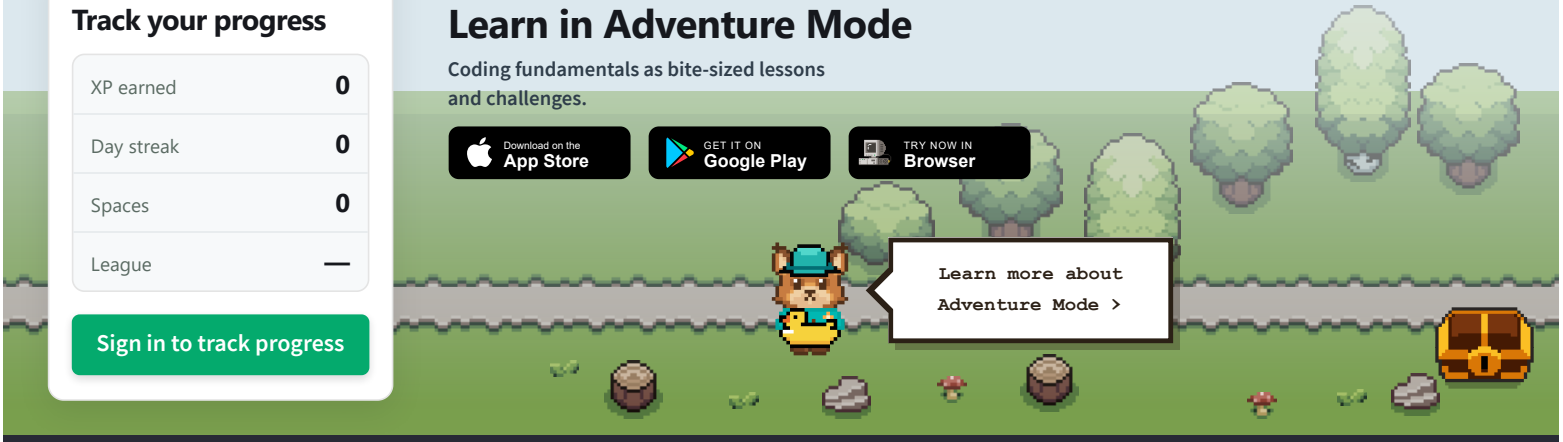
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String length

‹ Previous JavaScript String Reference Next ›

Example

```
let text = "Hello World!";
let length = text.length;
```

Try it Yourself »

Description

SHARE

The `length` property returns the length of a string.

The `length` property of an empty string is 0.

Syntax

```
string.length
```

Return Value

Type	Description
A number	The length of the string.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`length` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›

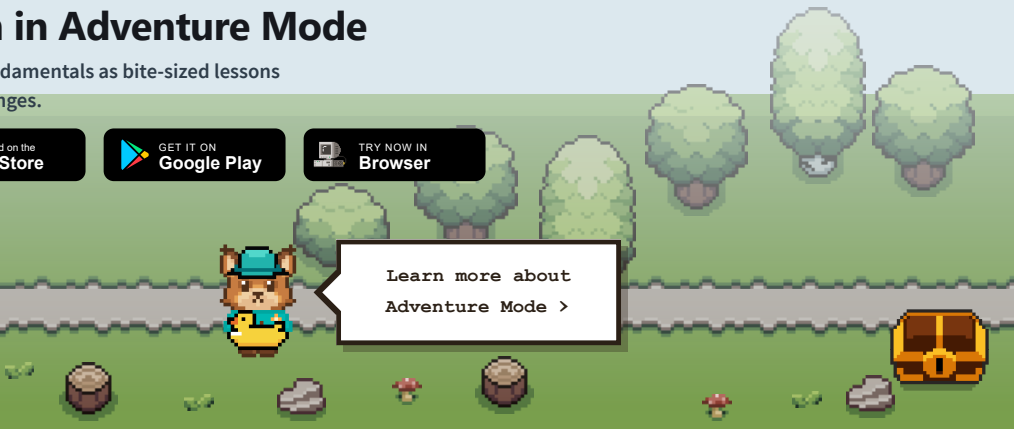


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String localeCompare()

‹ Previous JavaScript String Reference Next ›

Examples

Compare "ab" with "cd":

```
let text1 = "ab";
let text2 = "cd";
let result = text1.localeCompare(text2);
```

Try it Yourself »

```
let text1 = "cd";
let text2 = "ab";
let result = text1.localeCompare(text2);
```

Try it Yourself »

More examples below.

Description

SHARE

The `localeCompare()` method compares two strings in the current locale.

The `localeCompare()` method returns sort order -1, 1, or 0 (for before, after, or equal).

The current locale is based on the language settings of the browser.

Syntax

```
string.localeCompare(compareString)
```

Parameters

Parameter	Description
compareString	Required. The string to compare with.

Return Value

Type	Description
A number	One of 3 values: -1 if the string is sorted before the *compareString* 0 if the two strings are equal 1 if the string is sorted after the *compareString*

REMOVE ADS

More Examples

```
let text1 = "ab";
let text2 = "ab";
let result = text1.localeCompare(text2);
```

Try it Yourself »

```
let text1 = "A";
let text2 = "a";
let result = text1.localeCompare(text2);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`localeCompare()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›

Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

Learn in Adventure Mode

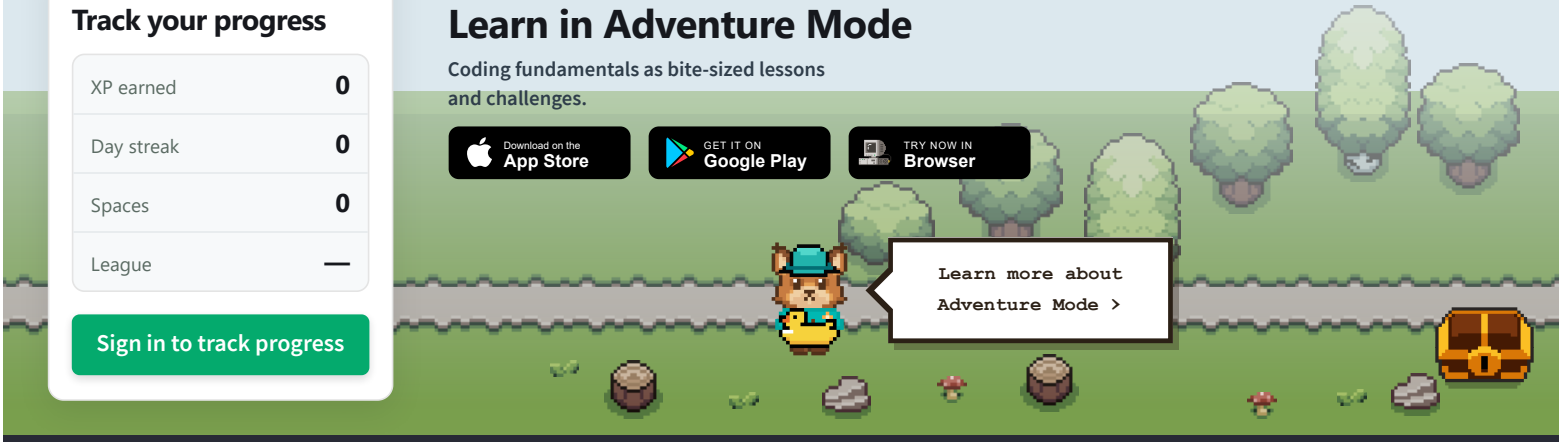
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser

Learn more about Adventure Mode >



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String match()

‹ Previous JavaScript String Reference Next ›

Examples

A search for "ain" using a string:

```
let text = "The rain in SPAIN stays mainly in the plain";
text.match("ain");
```

Try it Yourself »

A search for "ain" using a regular expression:

```
let text = "The rain in SPAIN stays mainly in the plain";
text.match(/ain/);
```

Try it Yourself »

A global search for "ain":

```
let text = "The rain in SPAIN stays mainly in the plain";
text.match(/ain/g);
```

Try it Yourself »

A global, case-insensitive search:

```
let text = "The rain in SPAIN stays mainly in the plain";
text.match(/ain/gi);
```

Try it Yourself »

Description

SHARE

The `match()` method matches a string against a regular expression **

The `match()` method returns an array with the matches.

The `match()` method returns *null* if no match is found.

Note

** If the search value is a string, it is converted to a regular expression.

See Also:

- String Match
- Regular Expression Tutorial
- Regular Expression Reference

Syntax

```
string.match(match)
```

Parameters

Parameter	Description
match	Required. The search value. A regular expression (or a string that will be converted to a regular expression).

Return Values

Type	Description
An array or `null`	An array containing the matches. `null` if no match is found.

REMOVE ADS

The Difference Between String match() and String search()

The `match()` method returns an array of matches.

The `search()` method returns the position of the first match.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Regular Expression Methods

Regular Expression **Search** and **Replace** can be done with different methods.

These are the most common:

String Methods

Method	Description
match(*regex*)	Returns an Array of results
matchAll(*regex*)	Returns an Iterator of results
replace(*regex*)	Returns a new String
replaceAll(*regex*)	Returns a new String
search(*regex*)	Returns the index of the first match
split(*regex*)	Returns an Array of results

RegExp Methods

Method	Description

regex.exec()	Returns a result Array or null
regex.test()	Returns true or false

Browser Support

`string.match()` is an ECMAScript1 (JavaScript 1997) feature.

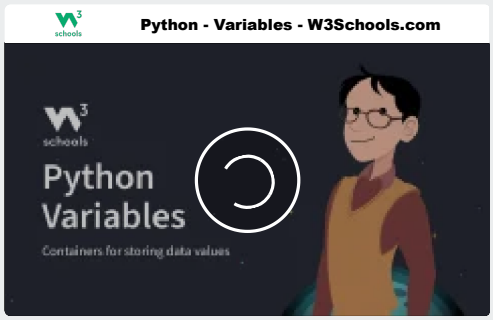
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript String Reference | Next ›

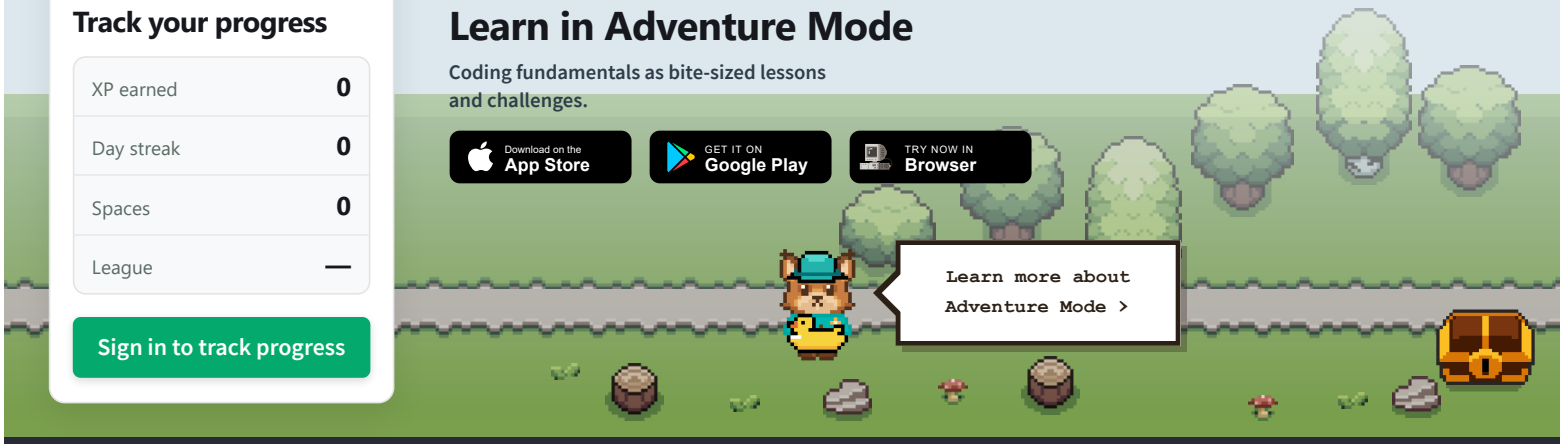


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



PLUS
SPACES
GET CERTIFIED
FOR TEACHERS
PRACTICE
CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String matchAll()

‹ Previous JavaScript String Reference Next ›

Before ES2020 there was no string method that could be used to search for all occurrences of a string in a string.

Example

```
const iterator = text.matchAll("Cats");
```

Try it Yourself »

If the parameter is a regular expression, the global flag (g) must be set set, otherwise a TypeError is thrown.

Example

```
const iterator = text.matchAll(/Cats/g);
```

Try it Yourself »

If you want to search case insensitive, the insensitive flag (i) must be set:

Example

```
const iterator = text.matchAll(/Cats/gi);
```

Try it Yourself »

Description

SHARE

The `matchAll()` method matches a string against a regular expression **

The `matchAll()` method returns an array with the matches.

The `matchAll()` method returns *null* if no match is found.

Note

** If the search value is a string, it is converted to a regular expression.

See Also:

- String Match
- Regular Expression Tutorial
- Regular Expression Reference

Syntax

```
string.matchAll(match)
```

Parameters

Parameter	Description
match	Required. The search value. A regular expression (or a string that will be converted to a regular expression).

Return Values

Type	Description
Iterator or `null`	An Iterator containing the matches. `null` if no match is found.

REMOVE ADS

The Difference Between String match() and String search()

The `match()` method returns an array of matches.

The `search()` method returns the position of the first match.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Regular Expression Methods

Regular Expression **Search** and **Replace** can be done with different methods.

These are the most common:

String Methods

Method	Description
match(*regex*)	Returns an Array of results
matchAll(*regex*)	Returns an Iterator of results
replace(*regex*)	Returns a new String
replaceAll(*regex*)	Returns a new String
search(*regex*)	Returns the index of the first match
split(*regex*)	Returns an Array of results

RegExp Methods

Method	Description

regex.exec()	Returns a result Array or null
regex.test()	Returns true or false

Browser Support

`string.matchAll()` is a JavaScript 2020 feature.

ES 2020 is supported in all modern browsers since **September 2020**:

Chrome	Edge	Firefox	Safari	Opera
85	85	79	14	71
Aug 2020	Aug 2020	Mar 2020	Sep 2020	Sep 2020

‹ Previous JavaScript String Reference Next ›

W3Schools
Get Certified in JavaScript
Boost your career with an official certification
$95
Get certified

REMOVE ADS

Python - Syntax - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

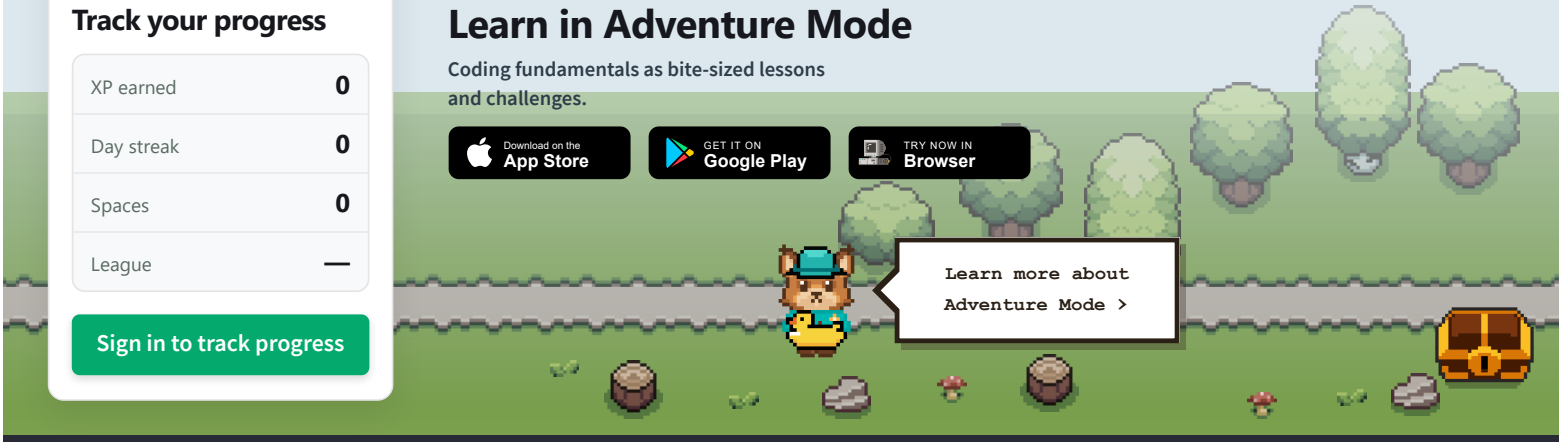
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String padEnd()

‹ Previous JavaScript String Reference Next ›

Examples

```
let text = "5";
let padded = text.padEnd(4,"0");
```

Try it Yourself »

```
let text = "5";
let padded = text.padEnd(4,"x");
```

Try it Yourself »

Description

SHARE

The `padEnd()` method pads a string at the end.

The `padEnd()` method pads a string with another string (multiple times) until it reaches a given length.

See Also:

The padStart() Method

The trim() Method

The trimEnd() Method

The trimStart() Method

Note

The `padEnd()` method is a string method.

To pad a number, convert the number to a string first.

See the example below.

Example

```
let numb = 5;
let text = numb.toString();
let padded = text.padEnd(4,"0");
```

Try it Yourself »

Syntax

```
string.padEnd(length, string)
```

Parameters

Parameter	Description
length	Required. The length of the resulting string.
string	Optional. The string to pad with. Default is space.

Return Value

Type	Description
A string	A String of the specified length, with the padding applied at the end.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

REMOVE ADS

Browser Support

`string.padEnd()` is a JaveScript 2017 feature.

ES 2017 is supported in all modern browsers since **September 2017**:

Chrome 58	Edge 15	Firefox 52	Safari 11	Opera 45
Apr 2017	Apr 2017	Mar 2017	Sep 2017	May 2017

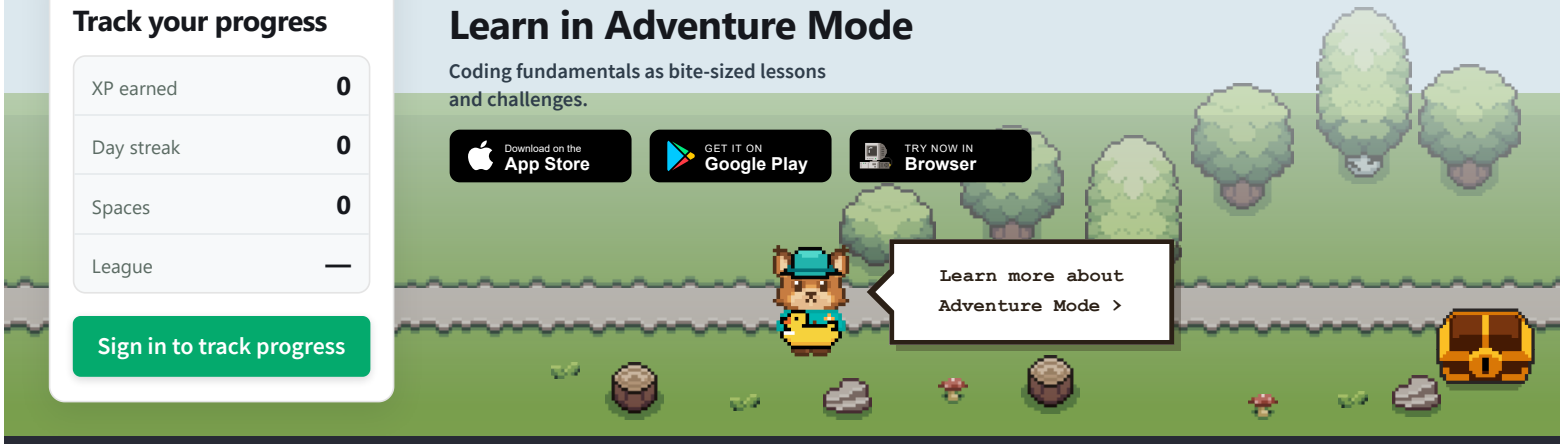
‹ Previous JavaScript String Reference Next ›

W3Schools
JS
Get Certified in JavaScript
Boost your career with an official certification
$95
Get certified

REMOVE ADS

FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.



JavaScript String padStart()

‹ Previous JavaScript String Reference Next ›

Examples

Pad a string with "0" until it reaches the length 4:

```
let text = "5";
let padded = text.padStart(4,"0");
```

Try it Yourself »

Pad a string with "x" until it reaches the length 4:

```
let text = "5";
let padded = text.padStart(4,"x");
```

Try it Yourself »

Description

SHARE

The `padStart()` method pads a string from the start.

The `padStart()` method pads a string with another string (multiple times) until it reaches a given length.

See Also:

The padEnd() Method

The trim() Method

The trimEnd() Method

The trimStart() Method

Note

The `padStart()` method is a string method.

To pad a number, convert the number to a string first.

See the example below.

Example

```
let numb = 5;
let text = numb.toString();
let padded = text.padStart(4,"0");
```

Try it Yourself »

Syntax

```
string.padStart(length, string)
```

Parameters

Parameter	Description
length	Required. The length of the resulting string.
string	Optional. The string to pad with. Default is space.

Return Value

Type	Description
A string	A String of the specified length, with the padding applied from the start.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

REMOVE ADS

Browser Support

`padStart()` is a JaveScript 2017 feature.

ES 2017 is supported in all modern browsers since **September 2017**:

				
Chrome 58	Edge 15	Firefox 52	Safari 11	Opera 45
Apr 2017	Apr 2017	Mar 2017	Sep 2017	May 2017

‹ Previous JavaScript String Reference Next ›

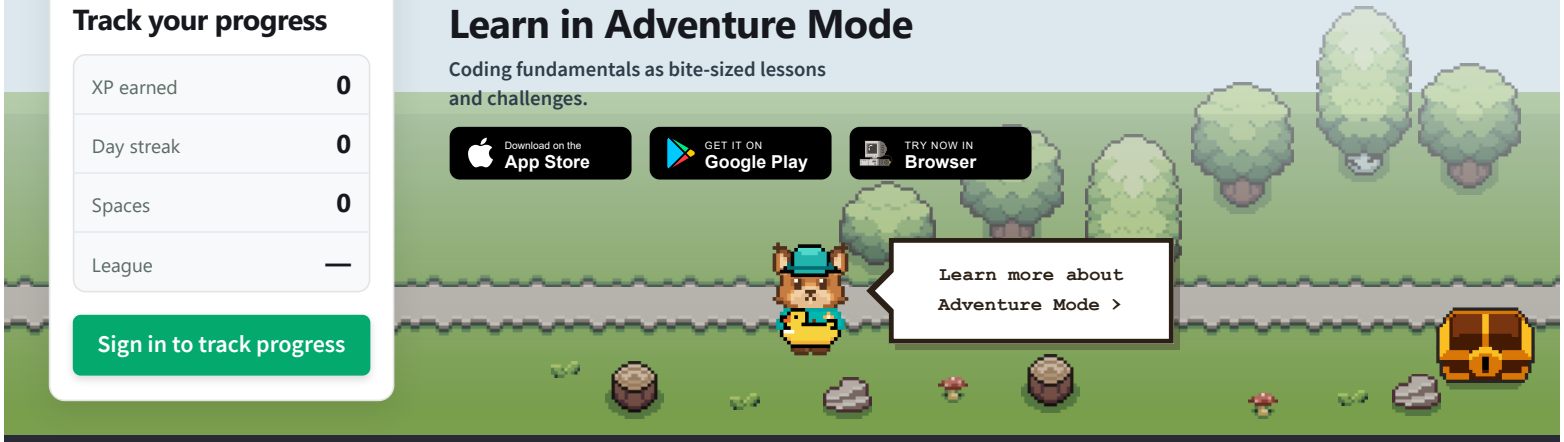


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



PLUS
SPACES
GET CERTIFIED
FOR TEACHERS
PRACTICE
CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial

W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String prototype

‹ Previous JavaScript String Reference Next ›

Example

Use the prototype property to add a new property to all objects of a given type:

```
function employee(name, jobtitle, born) {
  this.name = name;
  this.jobtitle = jobtitle;
  this.born = born;
}
employee.prototype.salary = 2000;

const fred = new employee("Fred Flintstone", "Caveman", 1970);
```

Try it Yourself »

Description

SHARE

The `prototype` is a property available with all JavaScript objects.

The `prototype` property allows you to add new properties and methods to strings.

Syntax

```
object.prototype.name = value
```

Warning

You are not advised to change the prototype of an object that you do not control.

You should not change the prototype of built in JavaScript datatypes like:

- Numbers
- Strings
- Arrays
- Dates
- Booleans
- Function
- Objects

Only change the prototype of your own objects.

The prototype Property

The JavaScript `prototype` property allows you to add new properties to objects:

Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.eyeColor = eyecolor;
}

Person.prototype.nationality = "English";
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`prototype` is an ECMAScript1 (JavaScript 1997) feature.

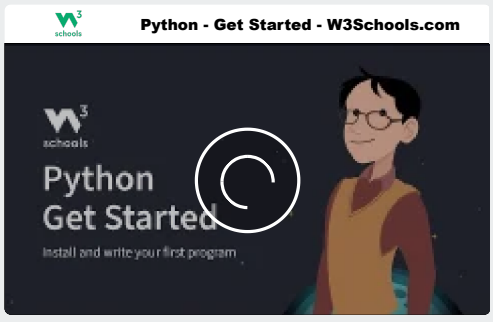
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS

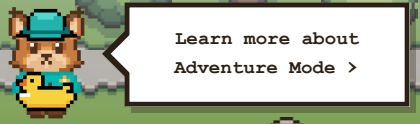


FREE ACCOUNT
Track your progress
XP earned 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Day streak 0
Spaces 0
League —
Sign in to track progress

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference

AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String repeat()

‹ Previous JavaScript String Reference Next ›

Examples

Create copies of a text:

```
let text = "Hello world!";
let result = text.repeat(2);
```

Try it Yourself »

```
let text = "Hello world!";
let result = text.repeat(4);
```

Try it Yourself »

Description

SHARE

The `repeat()` method returns a string with a number of copies of a string.

The `repeat()` method returns a new string.

The `repeat()` method does not change the original string.

Syntax

```
string.repeat(count)
```

Parameters

Parameter	Description
count	Required. The number of copies.

Return Value

Type	Description
A string	The copies of the original string.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`repeat()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

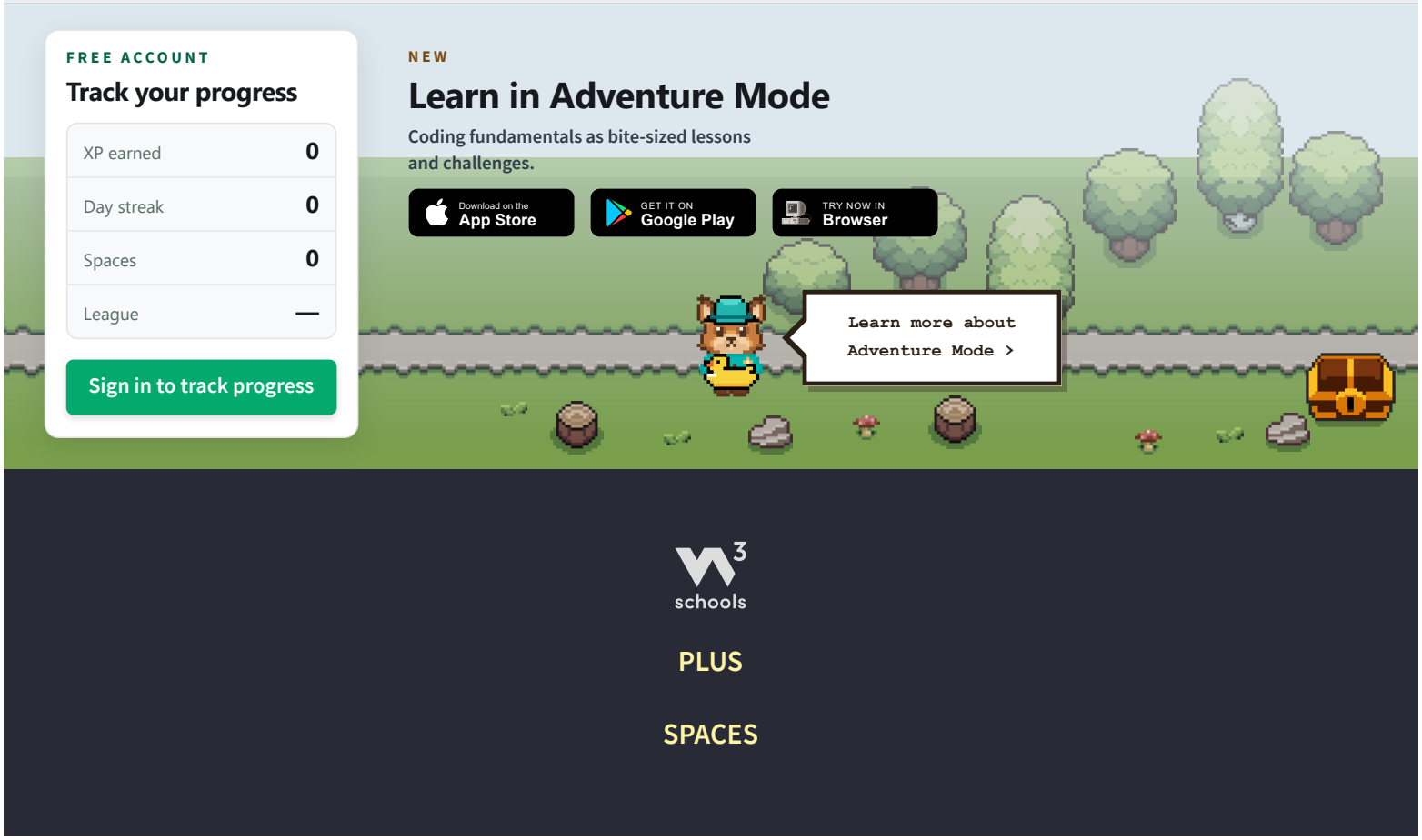
Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript String Reference Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String replace()

‹ Previous JavaScript String Reference Next ›

Examples

Replace Microsoft:

```
let text = "Visit Microsoft!";
let result = text.replace("Microsoft", "W3Schools");
```

Try it Yourself »

A global replacement:

```
let text = "Mr Blue has a blue house and a blue car";
let result = text.replace(/blue/g, "red");
```

Try it Yourself »

More examples below.

Description

SHARE

The `replace()` method searches a string for a value or a regular expression.

The `replace()` method returns a new string with the value(s) replaced.

The `replace()` method does not change the original string.

Note

If you replace a value, only the first instance will be replaced. To replace all instances, use a regular expression with the g modifier set.

Read more about regular expressions in our:

- RegExp Tutorial
- RegExp Reference

See Also:

The replaceAll() Method - replaces all matches

Syntax

```
string.replace(searchValue, newValue)
```

Parameters

Parameter	Description
searchValue	Required. The value, or regular expression, to search for.
newValue	Required. The new value (to replace with).

Return Value

Type	Description
A string	A new string where the specified value(s) has been replaced.

REMOVE ADS

More Examples

A global, case-insensitive replacement:

```
let text = "Mr Blue has a blue house and a blue car";
let result = text.replace(/blue/gi, "red");
```

Try it Yourself »

A function to return the replacement text:

```
let text = "Mr Blue has a blue house and a blue car";
let result = text.replace(/blue|house|car/gi, function (x) {
  return x.toUpperCase();
});
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`replace()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

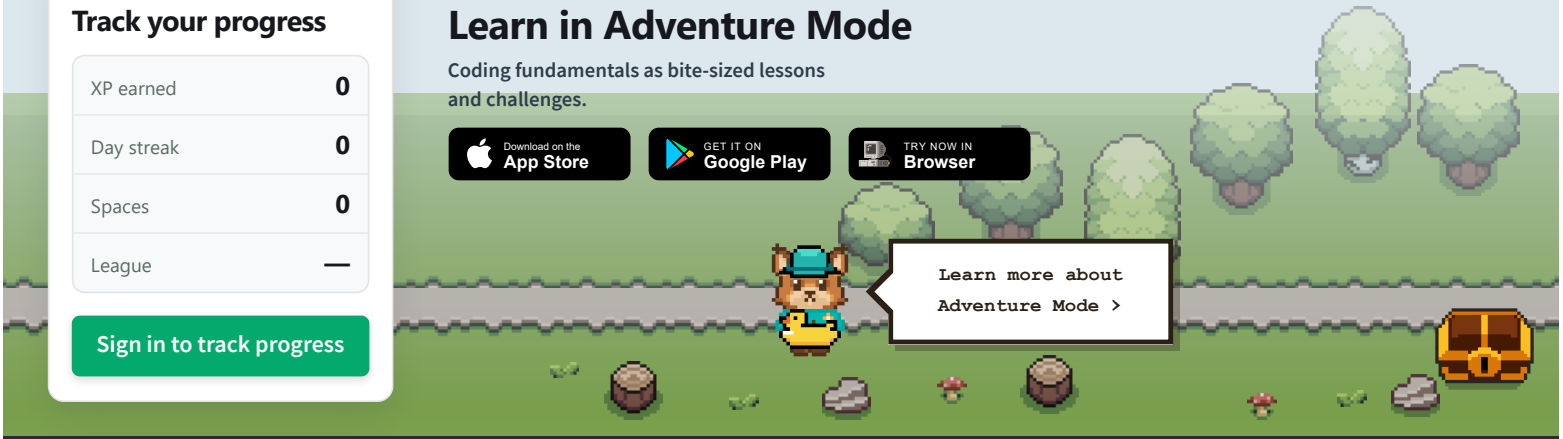
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String replaceAll()

‹ Previous JavaScript String Reference Next ›

Examples

```
text = text.replaceAll("Cats","Dogs");
text = text.replaceAll("cats","dogs");
```

Try it Yourself »

```
text = text.replaceAll(/Cats/g,"Dogs");
text = text.replaceAll(/cats/g,"dogs");
```

Try it Yourself »

More example below.

Description

SHARE

The `replaceAll()` method searches a string for a value or a regular expression.

The `replaceAll()` method returns a new string with all values replaced.

The `replaceAll()` method does not change the original string.

The `replaceAll()` method was introduced in JavaScript 2021.

Note

If the parameter is a regular expression, the global flag (g) must be set, otherwise a TypeError is thrown.

Read more about regular expressions in our:

- RegExp Tutorial
- RegExp Reference

Syntax

```
string.replaceAll(searchValue, newValue)
```

Parameters

Parameter	Description
searchValue	Required. The value, or regular expression, to search for.
newValue	Required. The new value (to replace with). This parameter can be a JavaScript function.

Return Value

Type	Description
A string	A new string where the search values has been replaced.

REMOVE ADS

More Examples

A global, case-insensitive replacement:

```
let text = "Mr Blue has a blue house and a blue car";
let result = text.replaceAll(/blue/gi, "red");
```

Try it Yourself »

A function to return the replacement text:

```
let text = "Mr Blue has a blue house and a blue car";
let result = text.replaceAll(/blue|house|car/gi, function (x) {
  return x.toUpperCase();
});
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

‹ Previous

JavaScript String Reference

Next ›



REMOVE ADS

Python - Comments - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

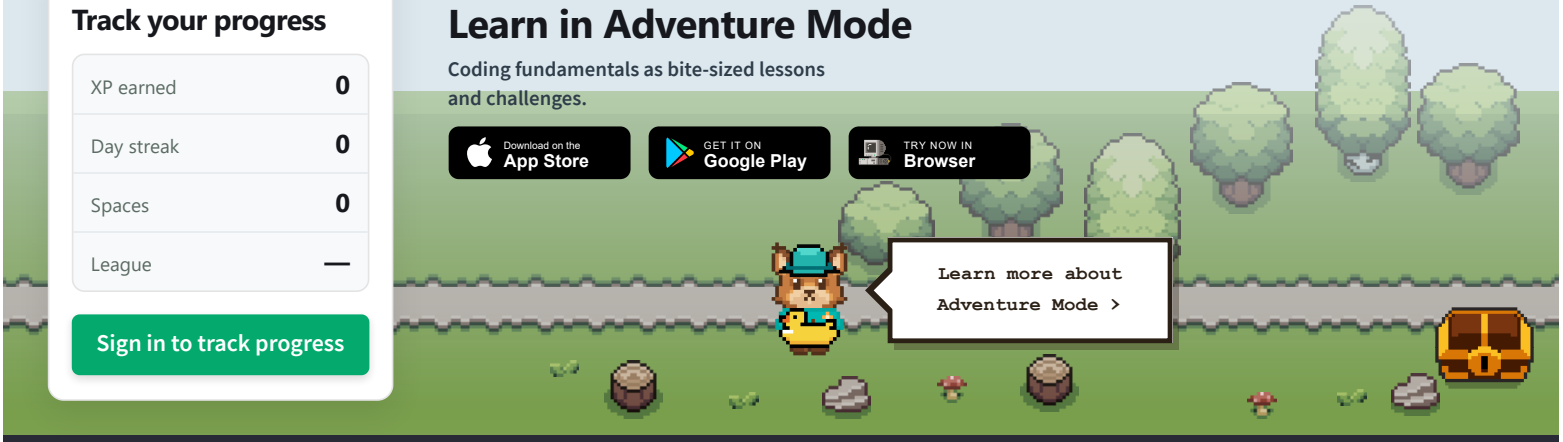
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String search()

‹ Previous JavaScript String Reference Next ›

Examples

Search for "Blue":

```
let text = "Mr. Blue has a blue house";
let position = text.search("Blue");
```

Try it Yourself »

Search for "blue":

```
let text = "Mr. Blue has a blue house";
let position = text.search("blue");
```

Try it Yourself »

Search for /Blue/:

```
let text = "Mr. Blue has a blue house";
let position = text.search(/Blue/);
```

Try it Yourself »

Search for /blue/:

```
let text = "Mr. Blue has a blue house";
let position = text.search(/blue/);
```

Try it Yourself »

Search case insensitive:

```
let text = "Mr. Blue has a blue house";
let position = text.search(/blue/i);
```

Try it Yourself »

Description

SHARE

The `search()` method matches a string against a regular expression **

The `search()` method returns the index (position) of the first match.

The `search()` method returns -1 if no match is found.

The `search()` method is case sensitive.

Note

** If the search value is a string, it is converted to a regular expression.

See Also:

- String Search
- Regular Expression Tutorial
- Regular Expression Reference

Syntax

```
string.search(searchValue)
```

Parameters

Parameter	Description
searchValue	Required. The search value. A regular expression (or a string that will be converted to a regular expression).

Return Value

Type	Description
A number	The position of the first match. -1 if no match.

REMOVE ADS

The Difference Between String search() and String indexOf()

The `search()` cannot take a start position argument.

The `indexOf()` method cannot search against a regular expression.

The Difference Between String search() and String match()

The `search()` method returns the position of the first match.

The `match()` method returns an array of matches.

Regular Expression Methods

Regular Expression **Search** and **Replace** can be done with different methods.

These are the most common:

String Methods

Method	**Description**
match(*regex*)	Returns an Array of results
matchAll(*regex*)	Returns an Iterator of results
replace(*regex*)	Returns a new String
replaceAll(*regex*)	Returns a new String
search(*regex*)	Returns the index of the first match

split(*regex*)	Returns an Array of results

RegExp Methods

Method	Description
regex.exec()	Returns a result Array or null
regex.test()	Returns true or false

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`search()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

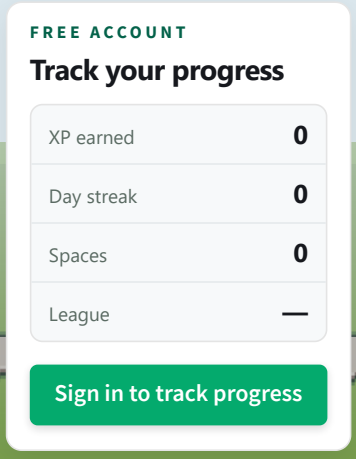
Chrome	Edge	Firefox	Safari	Opera

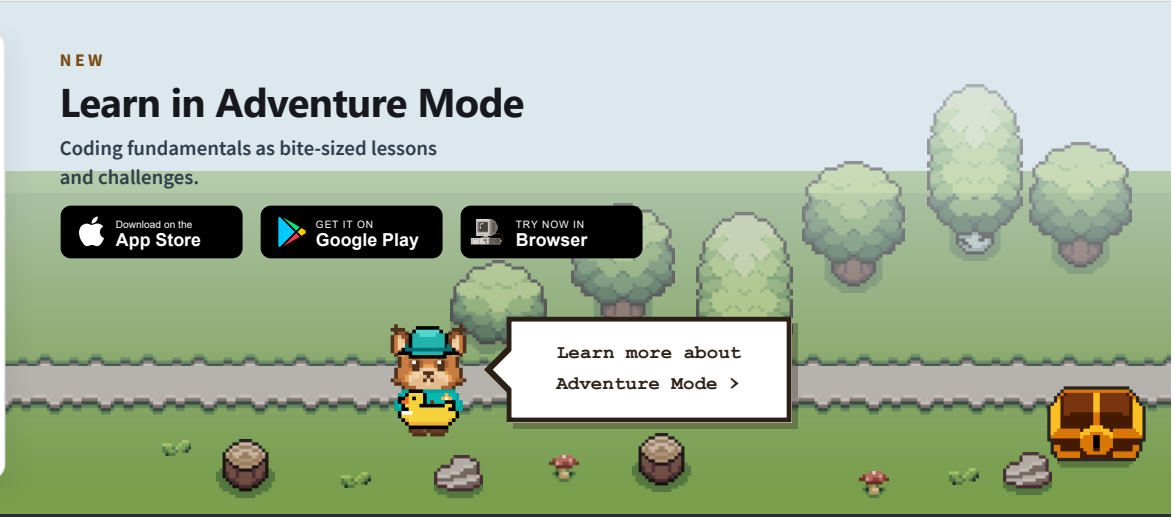
‹ Previous JavaScript String Reference Next ›



REMOVE ADS







schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String slice()

‹ Previous JavaScript String Reference Next ›

Examples

Slice the first 5 positions:

```
let text = "Hello world!";
let result = text.slice(0, 5);
```

Try it Yourself »

From position 3 to the end:

```
let result = text.slice(3);
```

Try it Yourself »

More examples below.

Description

SHARE

The `slice()` method extracts a part of a string.

The `slice()` method returns the extracted part in a new string.

The `slice()` method does not change the original string.

The start and end parameters specifies the part of the string to extract.

The first position is 0, the second is 1, ...

A negative number selects from the end of the string.

See Also

The split() Method

The substr() Method

The substring() Method

Syntax

```
string.slice(start, end)
```

Parameters

Parameter	Description
start	Required. The start position. (First character is 0).
end	Optional. The end position (up to, but not including). Default is string length.

Return Value

Type	Description
A string	The extracted part of the string.

REMOVE ADS

More Examples

From position 3 to 8:

```
let result = text.slice(3, 8);
```

Try it Yourself »

Only the first character:

```
let result = text.slice(0, 1);
```

Try it Yourself »

Only the last character:

```
let result = text.slice(-1);
```

Try it Yourself »

The whole string:

```
let result = text.slice(0);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`slice()` is an ECMAScript1 (JavaScript 1997) feature.

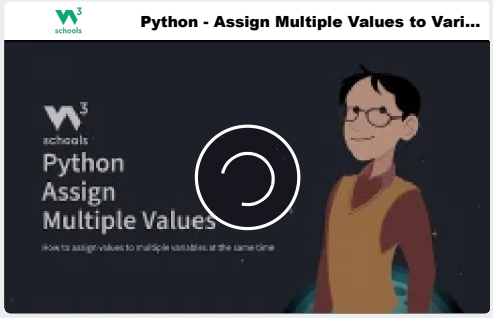
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String split()

‹ Previous JavaScript String Reference Next ›

Examples

Split the words:

```
let text = "How are you doing today?";
const myArray = text.split(" ");
```

Try it Yourself »

Split the words, and return the second word:

```
let text = "How are you doing today?";
const myArray = text.split(" ");
let word = myArray[1];
```

Try it Yourself »

Split the characters, including spaces:

```
const myArray = text.split("");
```

Try it Yourself »

Use the limit parameter:

```
const myArray = text.split(" ", 3);
```

Try it Yourself »

More examples below.

Description

SHARE

The `split()` method splits a string into an array of substrings.

The `split()` method returns the new array.

The `split()` method does not change the original string.

If (" ") is used as separator, the string is split between words.

See Also

The slice() Method

The substr() Method

The substring() Method

Syntax

```
string.split(separator, limit)
```

Parameters

Parameter	Description
separator	Optional. A string or regular expression to use for splitting. If omitted, an array with the original string is returned.
limit	Optional. An integer that limits the number of splits. Items after the limit are excluded.

Return Value

Type	Description

Array	An array containing the splitted values.

REMOVE ADS

More Examples

Split a string into characters and return the second character:

```
const myArray = text.split("");
```

Try it Yourself »

Use a letter as a separator:

```
const myArray = text.split("o");
```

Try it Yourself »

If the separator parameter is omitted, an array with the original string is returned:

```
const myArray = text.split();
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`split()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:







Chrome	Edge	Firefox	Safari	Opera

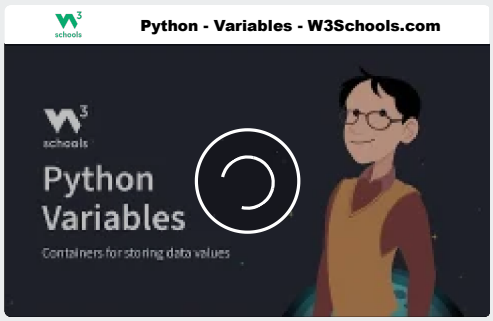
‹ Previous

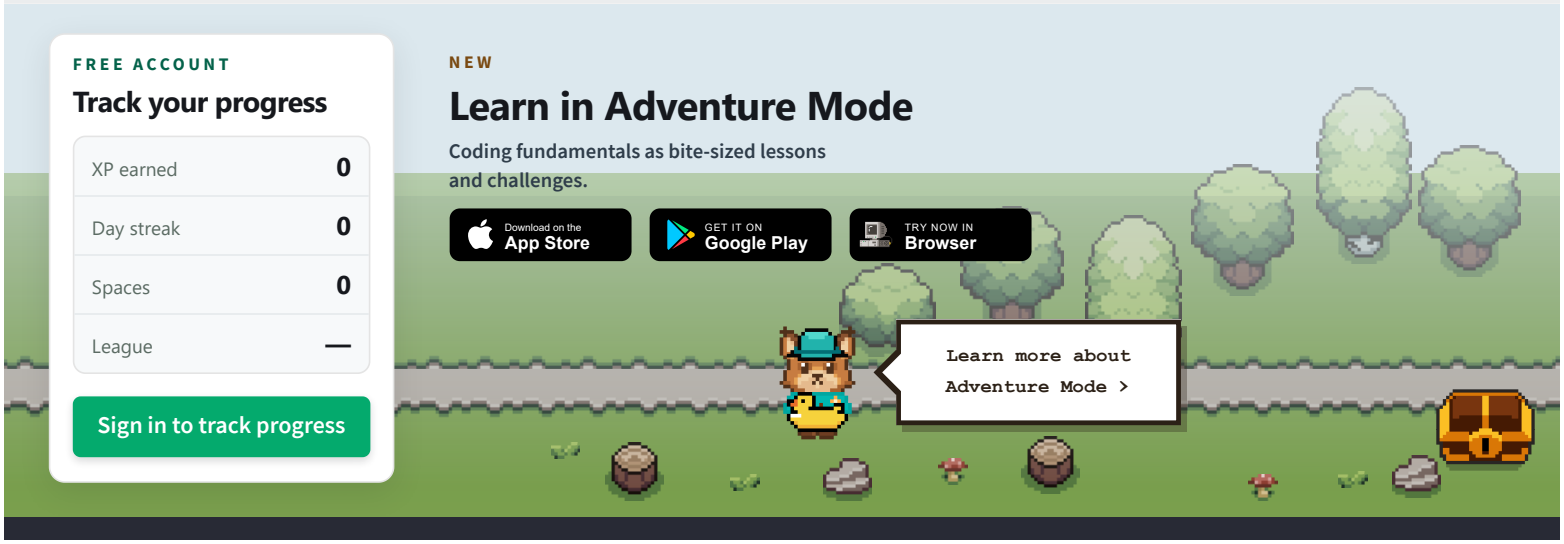
JavaScript String Reference

Next ›



REMOVE ADS





PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String startsWith()

‹ Previous JavaScript String Reference Next ›

Examples

Start at position 0 (true):

```
let text = "Hello world, welcome to the universe.";
text.startsWith("Hello");
```

Try it Yourself »

Start at position 1 (false):

```
let text = "Hello world, welcome to the universe.";
text.startsWith("Hello", 1);
```

Try it Yourself »

Description

SHARE

The `startsWith()` method returns `true` if a string starts with a specified string.

Otherwise it returns `false`.

The `startsWith()` method is case sensitive.

See Also:

The endsWith() Method

Syntax

```
string.startsWith(searchValue, start)
```

Parameters

Parameter	Description
searchValue	Required. The string to search for.
start	Optional. Start position. Default is 0.

Return Value

Type	Description
A boolean	Returns `true` if the string starts with the value. Otherwise it returns `false`.

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`startsWith()` is an ECMAScript6 (ES6 2015) feature.

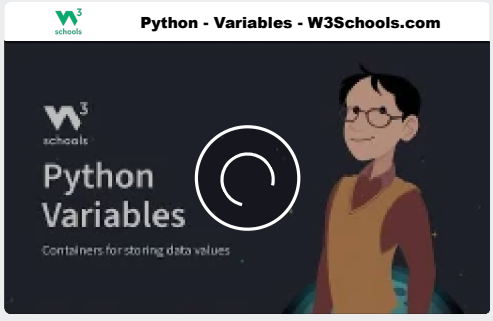
JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript String Reference | Next ›



REMOVE ADS

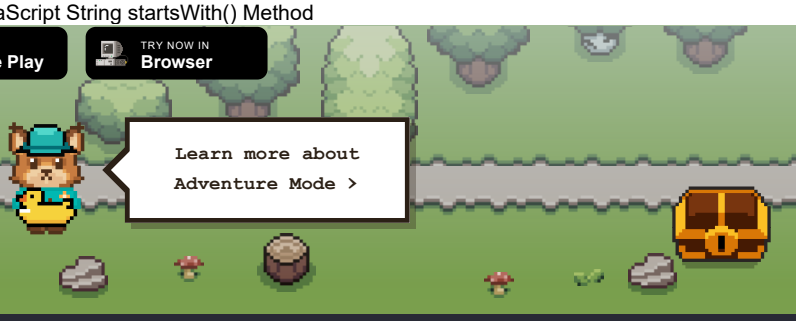


FREE ACCOUNT
Track your progress
XP earned 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Day streak 0
Spaces 0
League —
Sign in to track progress

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference

AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String substr()

‹ Previous JavaScript String Reference Next ›

Examples

Extract a substring from text:

```
let text = "Hello world!";
let result = text.substr(1, 4);
```

Try it Yourself »

Start at position 2:

```
let result = text.substr(2);
```

Try it Yourself »

More examples below.

Description

SHARE

The `substr()` method extracts a part of a string.

The `substr()` method begins at a specified position, and returns a specified number of characters.

The `substr()` method does not change the original string.

To extract characters from the end of the string, use a negative start position.

Warning

The `substr()` method is removed (deprecated) in the latest JavaScript standard.

Use `substring()` or `slice()` instead.

See Also:

The split() Method

The slice() Method

The substring() Method

Syntax

```
string.substr(start, length)
```

Parameters

Parameter	Description
start	Required. The start position. First character is at index 0. If *start* is greater than the length, substr() returns "". If *start* is negative, substr() counts from the end of the string.
length	Optional. The number of characters to extract. If omitted, it extracts the rest of the string

Return Value

Type	Description
A string	A string containing the extracted part. If *length* is 0 or negative, an empty string is returned.

REMOVE ADS

More Examples

Only the first:

```
let result = text.substr(0, 1);
```

Try it Yourself »

Only the last:

```
let result = text.substr(text.length-1, 1);
```

Try it Yourself »

The last 6:

```
let result = text.substr(-6, 6);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`substr()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript String Reference | Next ›

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

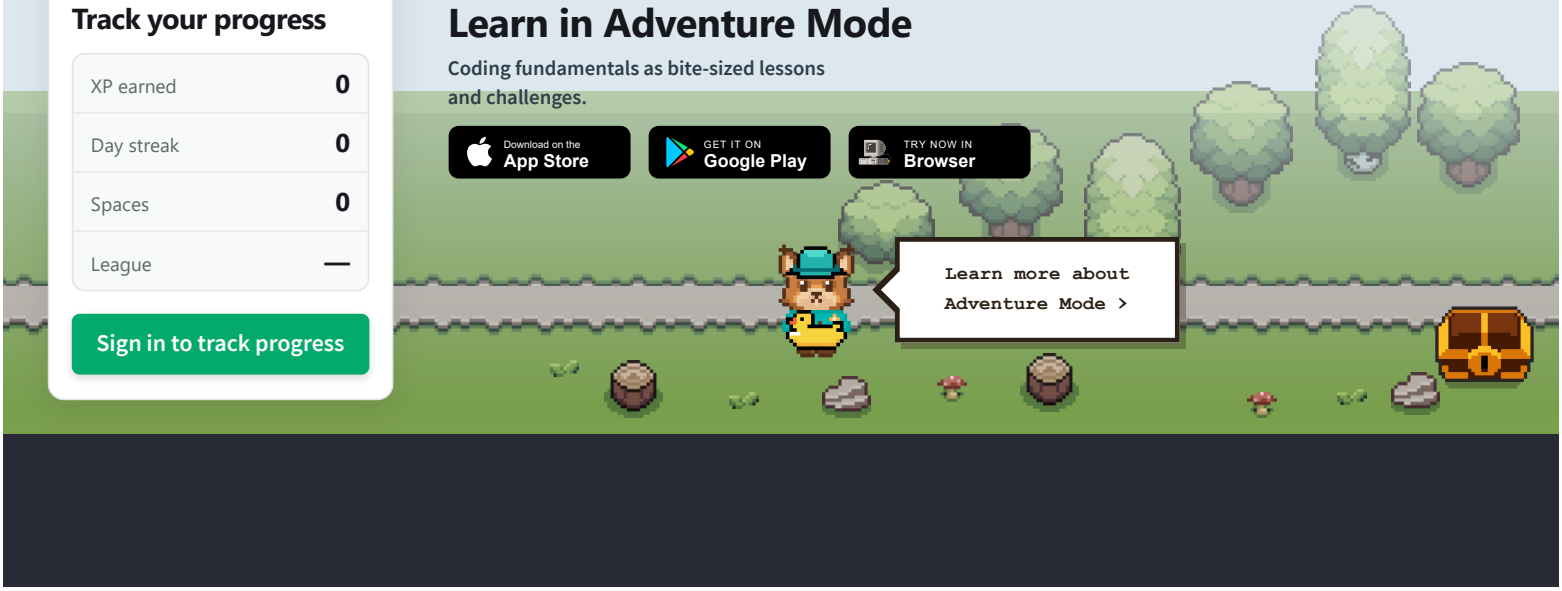
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String substring()

‹ Previous JavaScript String Reference Next ›

Examples

Extract a substring from text:

```
let text = "Hello world!";
let result = text.substring(1, 4);
```

Try it Yourself »

Start from position 2:

```
let result = text.substring(2);
```

Try it Yourself »

More examples below.

Description

SHARE

The `substring()` method extracts characters, between two indices (positions), from a string, and returns the substring.

The `substring()` method extracts characters from start to end (exclusive).

The `substring()` method does not change the original string.

If start is greater than end, arguments are swapped: (4, 1) = (1, 4).

Start or end values less than 0, are treated as 0.

See Also:

The split() Method

The slice() Method

The substr() Method

Syntax

```
string.substring(start, end)
```

Parameters

Parameter	Description
start	Required. Start position. First character is at index 0.
end	Optional. End position (up to, but not including). If omitted: the rest of the string.

Return Value

Type	Description
A string	A string containing the extracted characters.

REMOVE ADS

More Examples

If start is greater than end, parameters are swapped:

```
let result = text.substring(4, 1);
```

Try it Yourself »

If "start" is less than 0, it will start from index 0:

```
let result = text.substring(-3);
```

Try it Yourself »

Only the first:

```
let result = text.substring(0, 1);
```

Try it Yourself »

Only the last:

```
let result = text.substring(text.length - 1);
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`substring()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

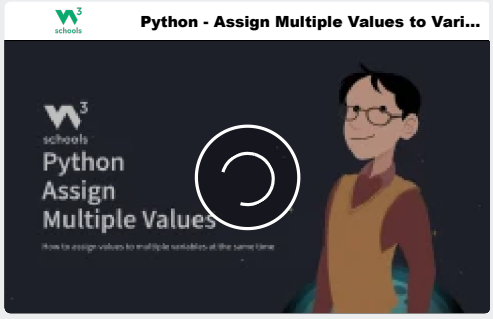
Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

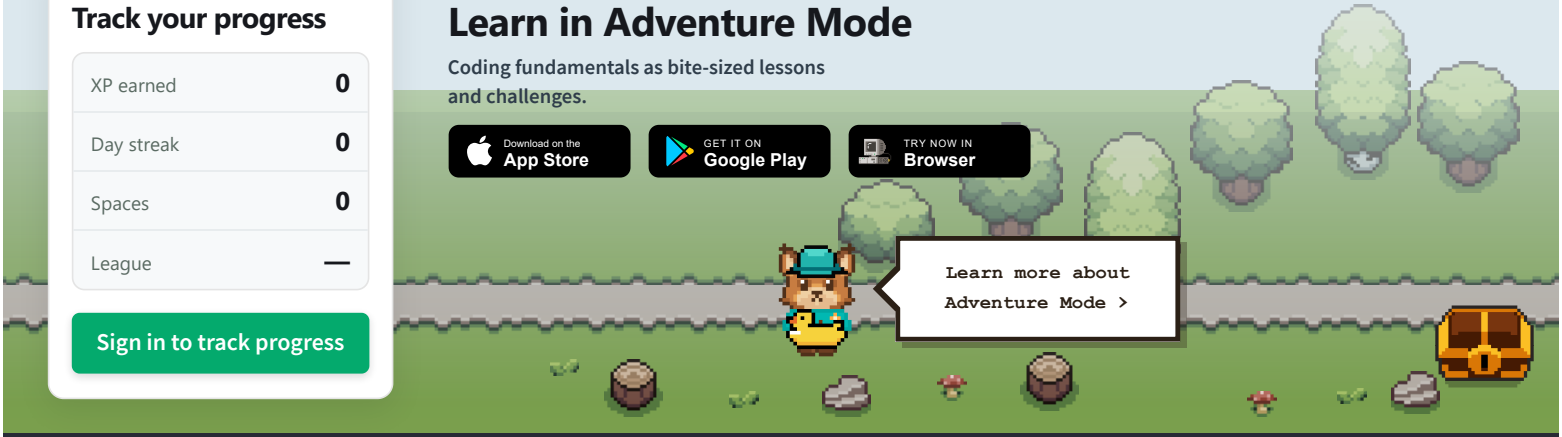
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String toLocaleLowerCase()

‹ Previous JavaScript String Reference Next ›

Example

Convert to lowercase:

```
let text = "Hello World!";
let result = text.toLocaleLowerCase();
```

Try it Yourself »

Description

SHARE

The `toLocaleLowerCase()` method converts a string to lowercase letters, using current locale.

The locale is based on the language settings of the browser.

The `toLocaleLowerCase()` method does not change the original string.

The `toLocaleLowerCase()` returns the same result as `toLowerCase()`, except for locales that conflict with the regular Unicode case mappings (such as Turkish).

See Also:

The toLowerCase() Method

The toUpperCase() Method

The toLocaleUpperCase() Method

Syntax

```
string.toLocaleLowerCase()
```

Parameters

NONE

Return Value

Type	Description
A string	A new string converted to lowercase according to current locale.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`toLocaleLowerCase()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

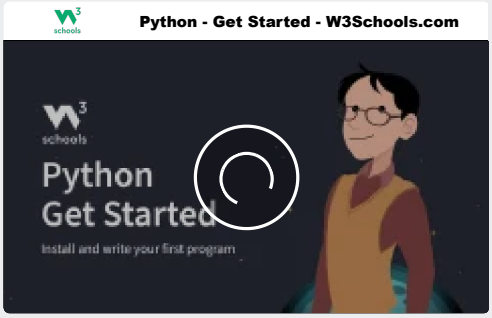
Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

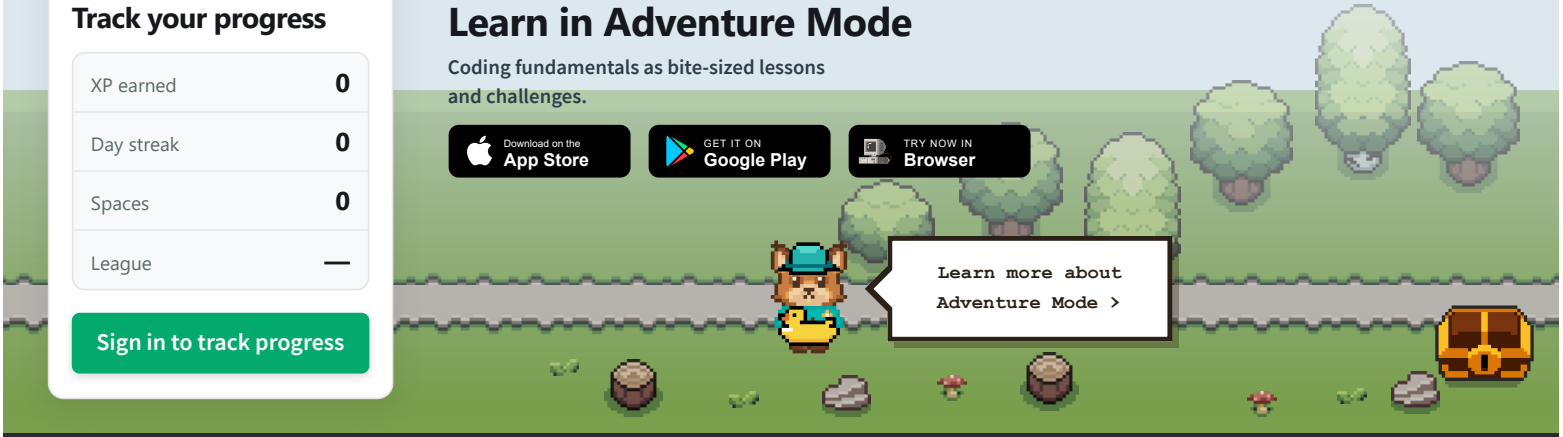
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String toLocaleUpperCase()

‹ Previous JavaScript String Reference Next ›

Example

Convert to uppercase:

```
let text = "Hello World!";
let result = text.toLocaleUpperCase();
```

Try it Yourself »

Description

SHARE

The `toLocaleUpperCase()` method converts a string to uppercase letters, using current locale.

The locale is based on the language settings of the browser.

The `toLocaleUpperCase()` method does not change the original string.

The `toLocaleUpperCase()` returns the same result as `toUpperCase()`, except for locales that conflict with the regular Unicode case mappings (such as Turkish).

See Also:

The toUpperCase() Method

The toLowerCase() Method

The toLocaleLowerCase() Method

Syntax

```
string.toLocaleUpperCase()
```

Parameters

NONE

Return Value

Type	Description
A string.	A new string converted to uppercase according to current locale.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

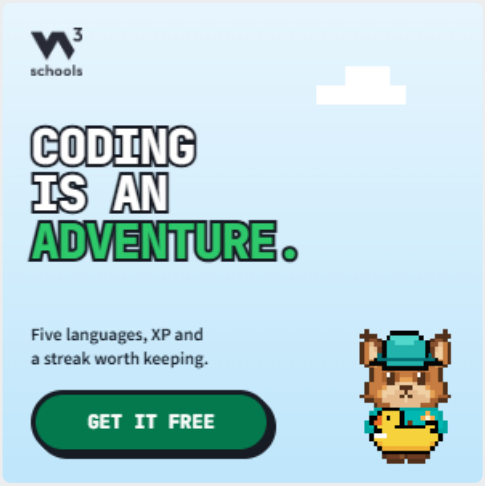
Browser Support

`toLocalUpperCase()` is an ECMAScript1 (JavaScript 1997) feature.

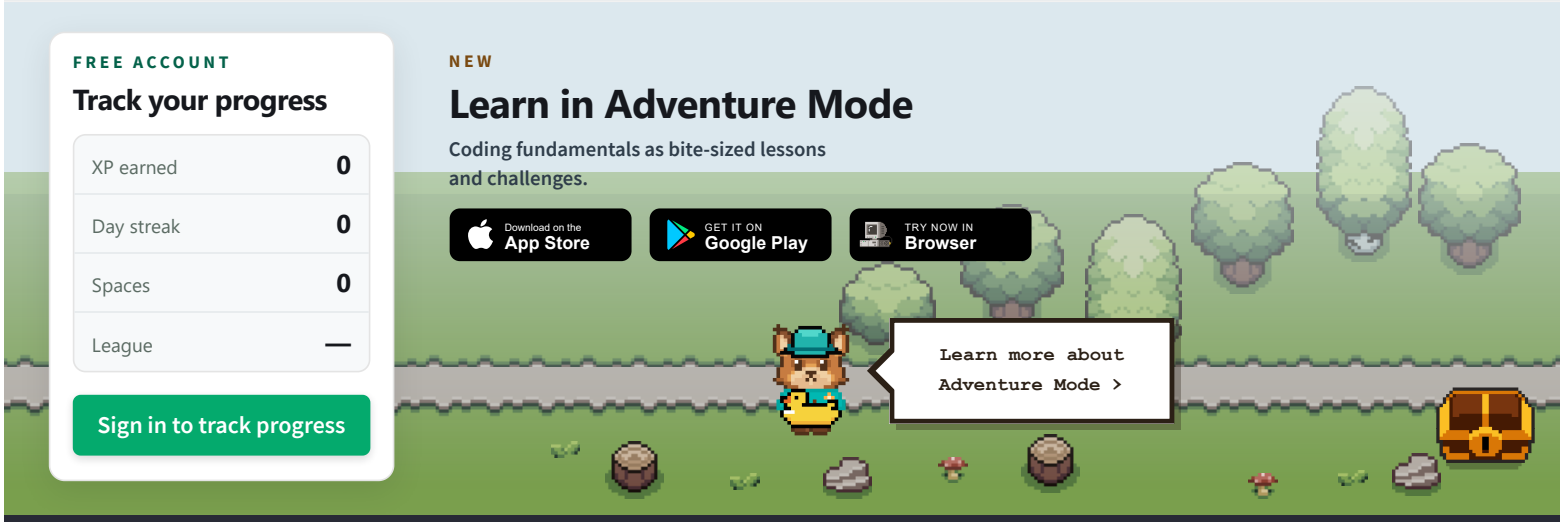
It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›



REMOVE ADS



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate

Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String toLowerCase()

‹ Previous JavaScript String Reference Next ›

Example

Convert to lowercase:

```
let text = "Hello World!";
let result = text.toLowerCase();
```

Try it Yourself »

Description

SHARE

The `toLowerCase()` method converts a string to lowercase letters.

The `toLowerCase()` method does not change the original string.

See Also:

The toUpperCase() Method

The toLocaleLowerCase() Method

The toLocaleUpperCase() Method

Syntax

```
string.toLowerCase()
```

Parameters

NONE

Return Value

Type	Description
A string	The string converted to lowercase.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`toLowerCase()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String toString()

‹ Previous JavaScript String Reference Next ›

Example

Get the content of the text:

```
let text = "Hello World!";
let result = text.toString();
```

Try it Yourself »

Same as:

```
let text = "Hello World!";
let result = text;
```

Try it Yourself »

More examples below.

Description

SHARE

The `toString()` method returns a string as a string.

The `toString()` method does not change the original string.

The `toString()` method can be used to convert a string object into a string.

See Also:

The valueOf() Method

Note

Every JavaScript object has a `toString()` method.

The `toString()` method is used by JavaScript when an object needs to be displayed as a text (like in HTML), or when an object needs to be used as a string.

Syntax

```
string.toString()
```

Parameters

NONE

Return Value

Type	Description
A string	The content of the string.

REMOVE ADS

More Examples

Get the value of a String object:

```
let text = new String("Hello World!");
let result = text.toString();
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`toString()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›



REMOVE ADS

FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.



JavaScript String toUpperCase()

‹ Previous JavaScript String Reference Next ›

Example

Convert to uppercase:

```
let text = "Hello World!";
let result = text.toUpperCase();
```

Try it Yourself »

Description

SHARE

The `toUpperCase()` method converts a string to uppercase letters.

The `toUpperCase()` method does not change the original string.

See Also:

The toLowerCase() Method

The toLocaleLowerCase() Method

The toLocaleUpperCase() Method

Syntax

```
string.toUpperCase()
```

Parameters

NONE

Return Value

Type	Description
A string	The string converted to uppercase.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`toUpperCase()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript String Reference Next ›

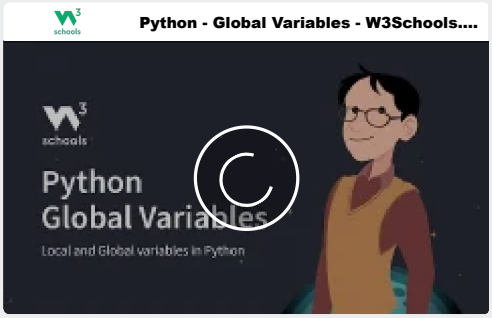
Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

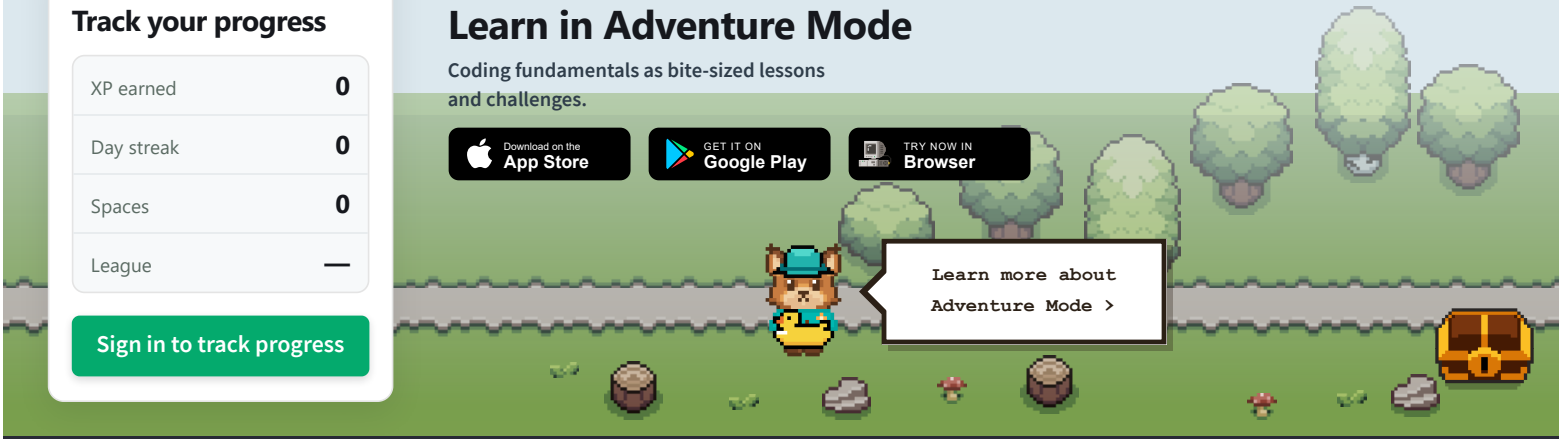
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String toWellFormed()

‹ Previous JavaScript String Reference Next ›

Examples

```
let text = "Hello World \uD800";
let result = text.toWellFormed();
```

Try it Yourself »

Description

SHARE

The `toWellFormed()` method returns returns a new string where all "lone surrogates" are replaced with the Unicode replacement character (U+FFFD).

The `toWellFormed()` method does not change the orginal string.

Lone Surrogates

A **Lone surrogate** is a Unicode surrogate code that is not a part of a valid surrogate pair, used to represent characters in UTF-16 encoding.

Syntax

```
string.toWellFormed()
```

Parameters

NONE

Return Value

Type	Description
String	A string where all "lone surrogates" are replaced with the Unicode replacement character (U+FFFD).

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

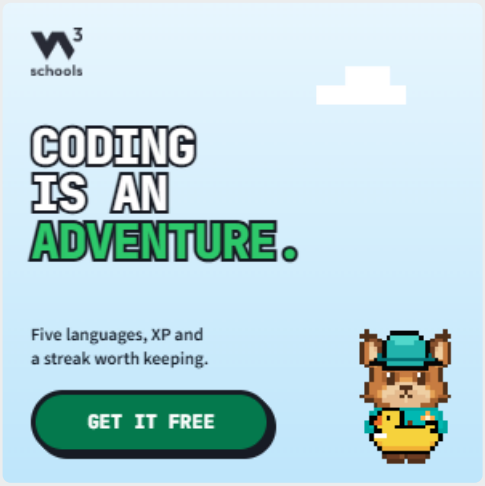
Browser Support

`toWellFormed()` is a JavaScrip 2024 feature.

ES 2024 is supported in all modern browsers since **March 2024**:

Chrome	Edge	Firefox	Safari	Opera
117	117	119	17.4	103
Sep 2023	Sep 2023	Oct 2023	Okt 2024	May 2023

‹ Previous JavaScript String Reference Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String trim()

‹ Previous JavaScript String Reference Next ›

Example 1

Remove spaces with trim():

```
let text = "       Hello World!        ";
let result = text.trim();
```

Try it Yourself »

Remove spaces with replace() using a regular expression:

```
let text = "       Hello World!        ";
let result = text.replace(/^\s+|\s+$/gm,'');
```

Try it Yourself »

Description

SHARE

The `trim()` method removes whitespace from both sides of a string.

The `trim()` method does not change the original string.

See Also:

The trimEnd() Method

The trimStart() Method

The padEnd() Method

The padStart() Method

Syntax

```
string.trim()
```

Parameters

NONE

Return Value

Type	Description
A string	A string with removed whitespace from both ends.

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`trim()` is an ECMAScript5 (ES5 2009) feature.

JavaScript 2009 is supported in all browsers since **July 2013**:





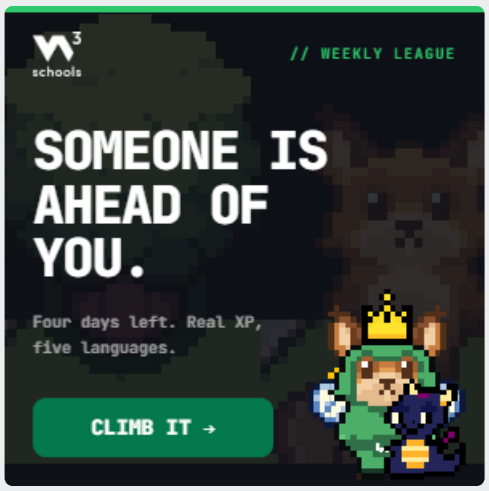


Chrome 23	IE/Edge 11	Firefox 21	Safari 6	Opera 15
Sep 2012	Sep 2012	Apr 2013	Jul 2012	Jul 2013

‹ Previous

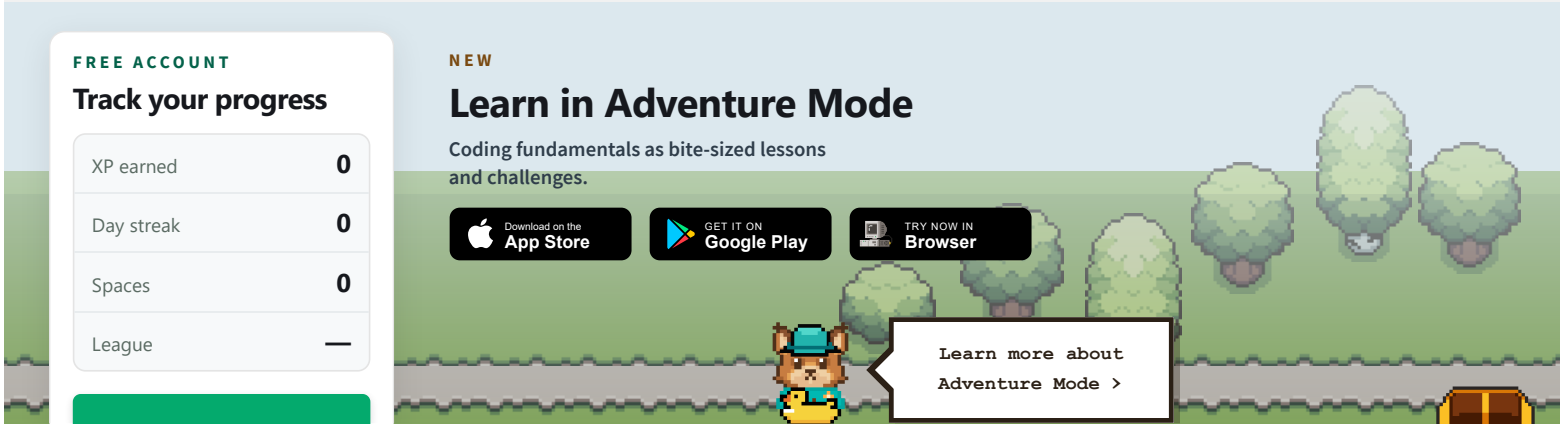
JavaScript String Reference

Next ›

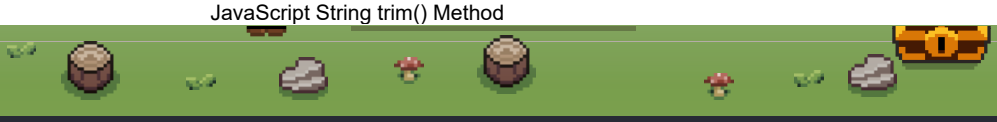


REMOVE ADS





Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String trimEnd()

‹ Previous JavaScript String Reference Next ›

Example

```
let text1 = "     Hello World!     ";
let text2 = text1.trimEnd();
```

Try it Yourself »

Description

SHARE

The `trimEnd()` method removes whitespace from the end of a string.

The `trimEnd()` method does not change the original string.

The `trimEnd()` method works like `trim()`, but removes whitespace only from the end of a string.

Note

The `trimEnd()` method was added to JavaScript in ECMAScript 2019.

See Also:

The trim() Method

The trimStart() Method

The padEnd() Method

The padStart() Method

Syntax

```
string.trimEnd()
```

Parameters

NONE

Return Value

Type	Description
A string	A string with removed whitespace from the end of the string.

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

JavaScript String `trimEnd()` is supported in all browsers since January 2020:

				
Chrome 66	Edge 79	Firefox 61	Safari 12	Opera 50
Apr 2018	Jan 2020	Jun 2018	Sep 2018	May 2018

‹ Previous | JavaScript String Reference | Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

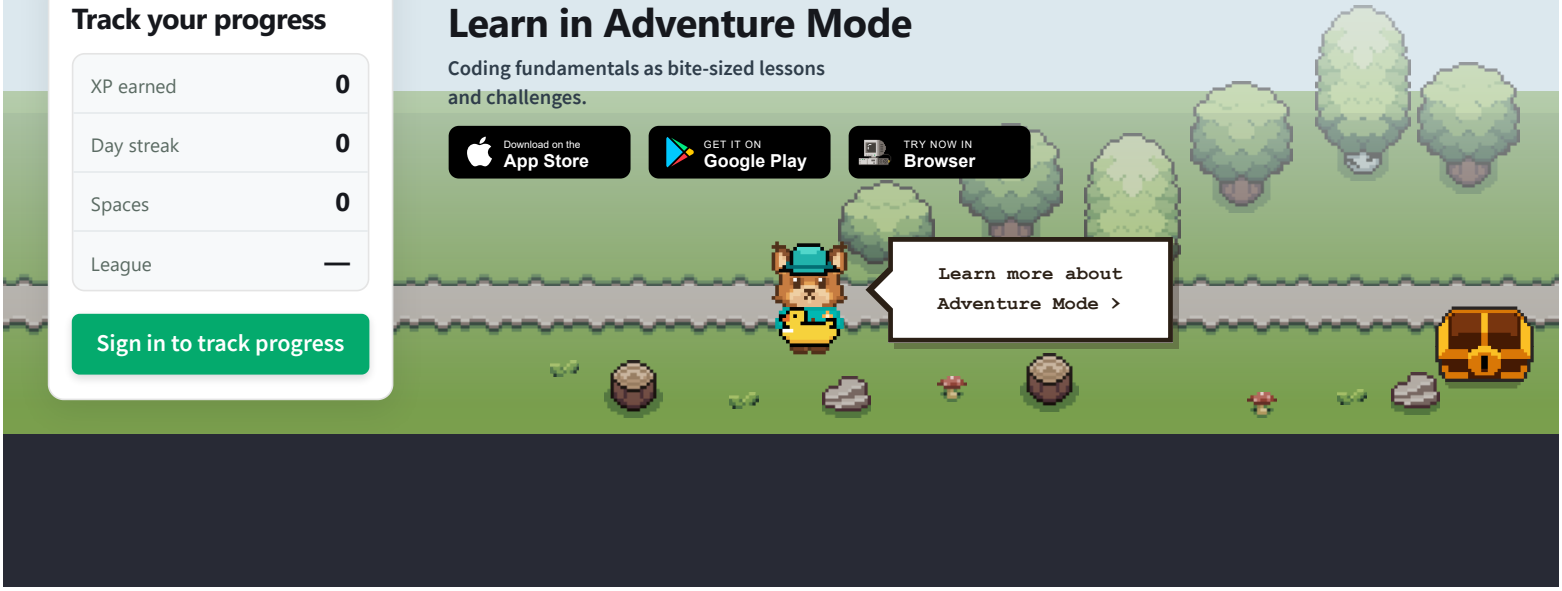
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String trimStart()

‹ Previous JavaScript String Reference Next ›

Example

```
let text1 = "     Hello World!     ";
let text2 = text1.trimStart();
```

Try it Yourself »

Description

SHARE

The `trimStart()` method removes whitespace from the beginning of a string.

The `trimStart()` method does not change the original string.

The `trimStart()` method works like `trim()`, but removes whitespace only from the start of a string.

Note

The `trimStart()` method was added to JavaScript in ECMAScript 2019.

See Also:

The trim() Method

The trimEnd() Method

The padEnd() Method

The padStart() Method

Syntax

```
string.trimStart()
```

Parameters

NONE

Return Value

Type	Description
A string	A string with removed whitespace from the beginning of the string.

REMOVE ADS

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

JavaScript String `trimStart()` is supported in all browsers since January 2020:

Chrome 66	Edge 79	Firefox 61	Safari 12	Opera 50
Apr 2018	Jan 2020	Jun 2018	Sep 2018	May 2018

‹ Previous | JavaScript String Reference | Next ›

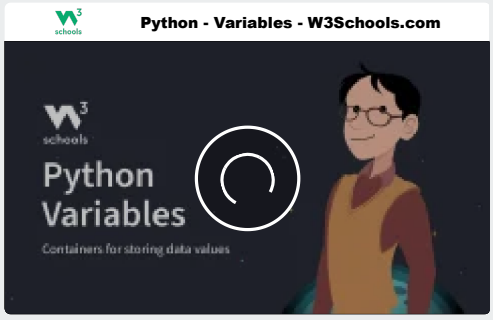
Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

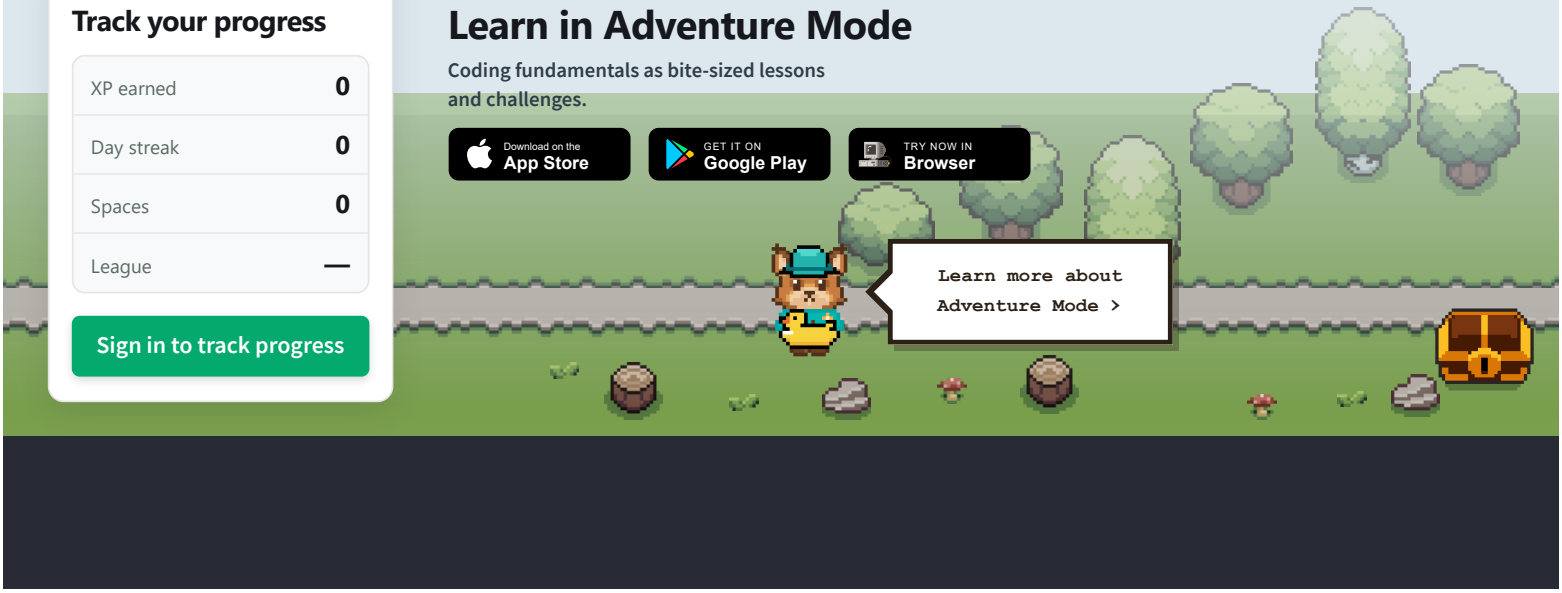
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String valueOf()

‹ Previous JavaScript String Reference Next ›

Example

Get the value of a text:

```
let text = "Hello World!";
let result = text.valueOf();
```

Try it Yourself »

The same as:

```
let text = "Hello World!";
let result = text;
```

Try it Yourself »

More examples below.

Description

SHARE

The `valueOf()` method returns the primitive value of a string.

The `valueOf()` method does not change the original string.

The `valueOf()` method can be used to convert a string object into a string.

See Also:

The toString() Method

Note

The `valueOf()` method is the default method for JavaScript strings.

It is used internally by JavaScript.

Normally, you will not use it in your code.

Syntax

```
string.valueOf()
```

Parameters

NONE

Return Value

Type	Description
A string	The primitive value of the string.

REMOVE ADS

More Examples

Get the value of a String object:

```
let text = new String("Hello World!");
let result = text.valueOf();
```

Try it Yourself »

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

`valueOf()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript String Reference

Next ›



REMOVE ADS

Python - Get Started - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

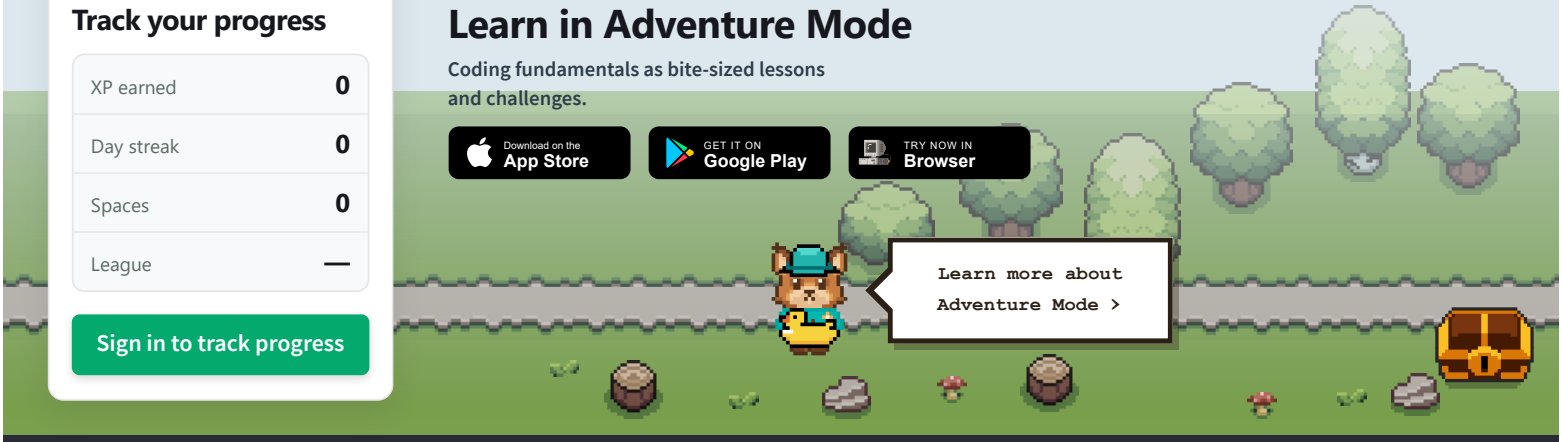
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String anchor()

‹ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.anchor("Chapter 10");
```

Try it Yourself »

Description

String anchor() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `anchor` method returns a string embedded in an <a> tag:

string

Syntax

```
string.anchor("name")
```

Parameters

Parameter	Description
name	Required. The name of the anchor.

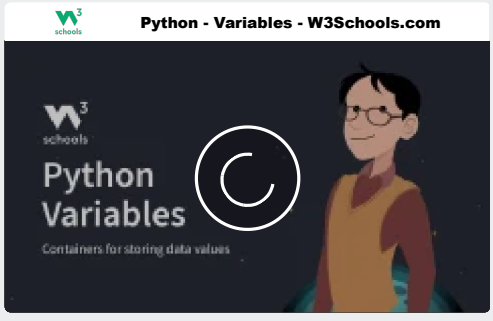
Return Value

A string embedded in an <a> tag.

❮ JavaScript String Reference

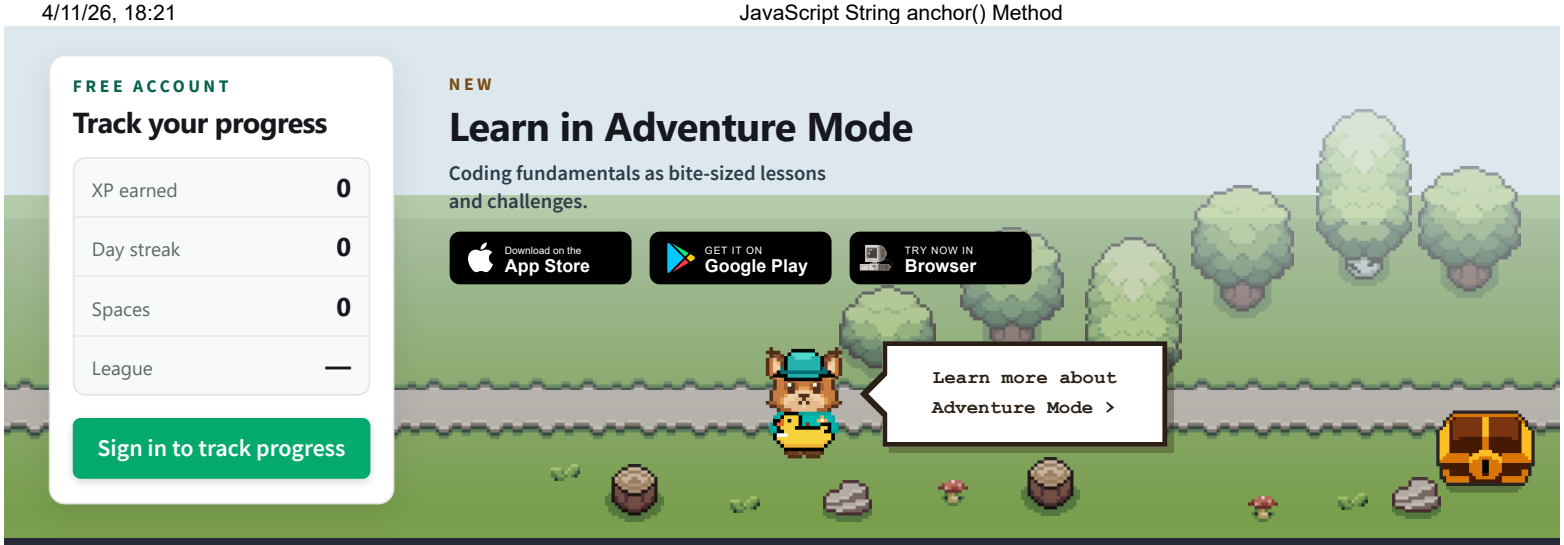


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String big()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.big();
```

Try it Yourself »

Description

String big() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `big()` method returns a string embedded in a <big> tag:

<big>string</big>

The <big> tag is not supported in HTML5.

Syntax

```
string.big()
```

Parameters

NONE

Return Value

A string embedded in a <big> tag.

‹ JavaScript String Reference

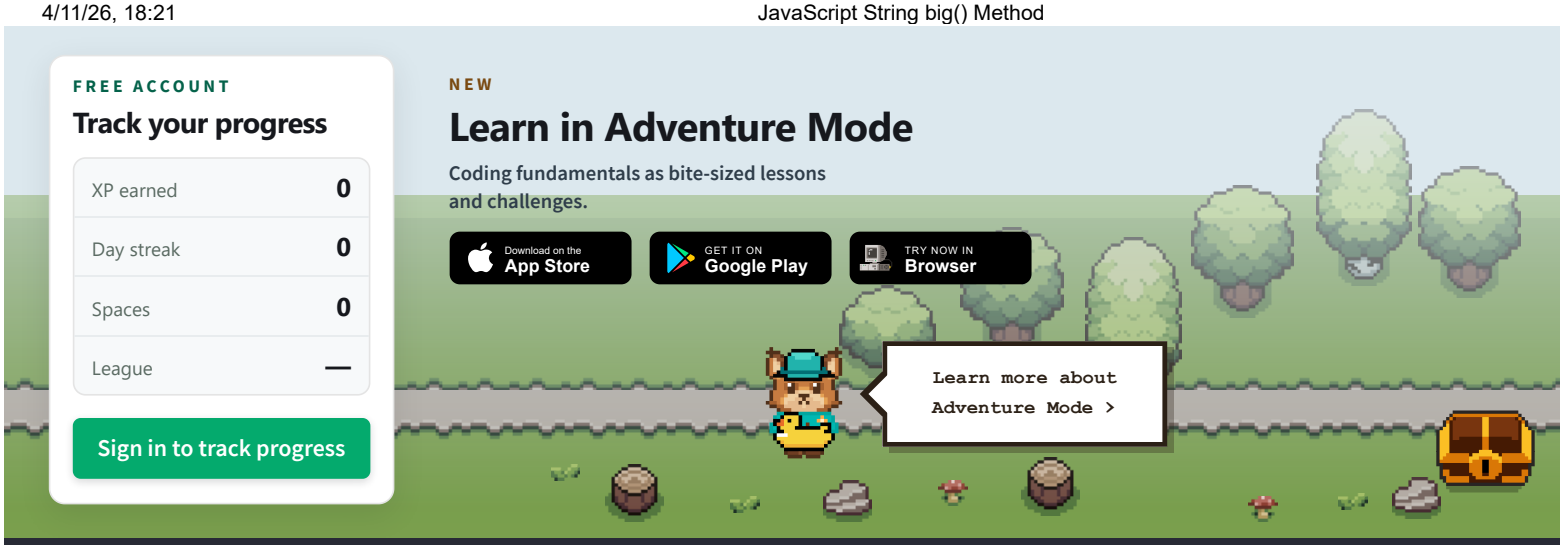


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >



PLUS
SPACES
GET CERTIFIED
FOR TEACHERS
PRACTICE
CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String blink()

‹ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.blink();
```

Try it Yourself »

Description

String blink() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `blink()` method a string embedded in a <blink> tag:

<blink>string</blink>

The <blink> tag is not supportet in HTML 5.

Browser Support

Chrome	IE	Edge	Firefox	Safari	Opera
NO	NO	NO	NO	NO	NO

Syntax

```
string.blink()
```

Parameters

NONE

Return Value

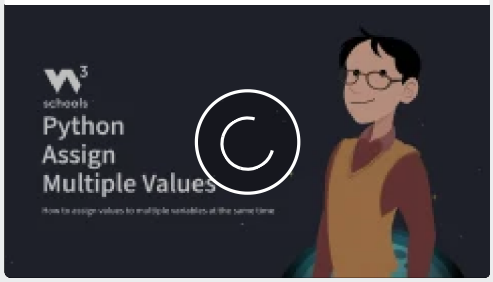
A string embedded in a <blink> tag.

‹ JavaScript String Reference



REMOVE ADS

Python - Assign Multiple Values to Vari...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

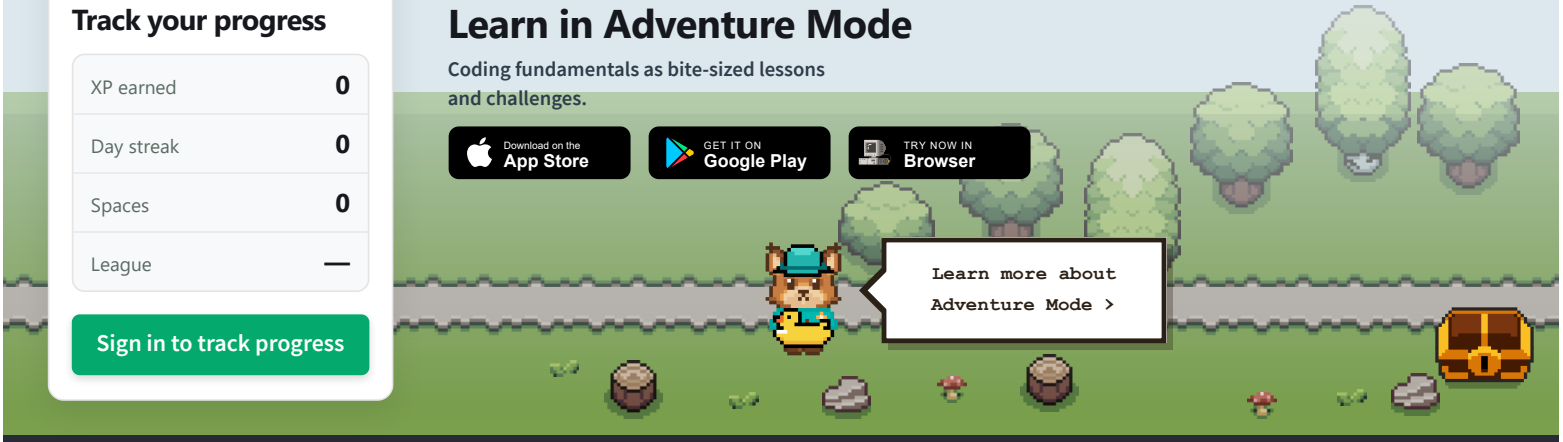
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String bold()

‹ JavaScript String Reference

Example

Display a text in bold:

```
let text = "Hello World!";
let result = text.bold();
```

Try it Yourself »

Description

String bold() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `bold()` method returns a string embedded in a tag:

string

Syntax

```
string.bold()
```

Parameters

NONE

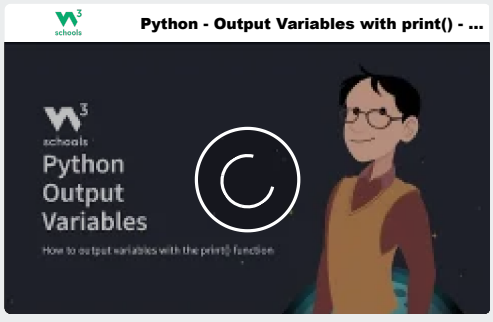
Return Value

A string embedded in a <b> tag.

‹ JavaScript String Reference

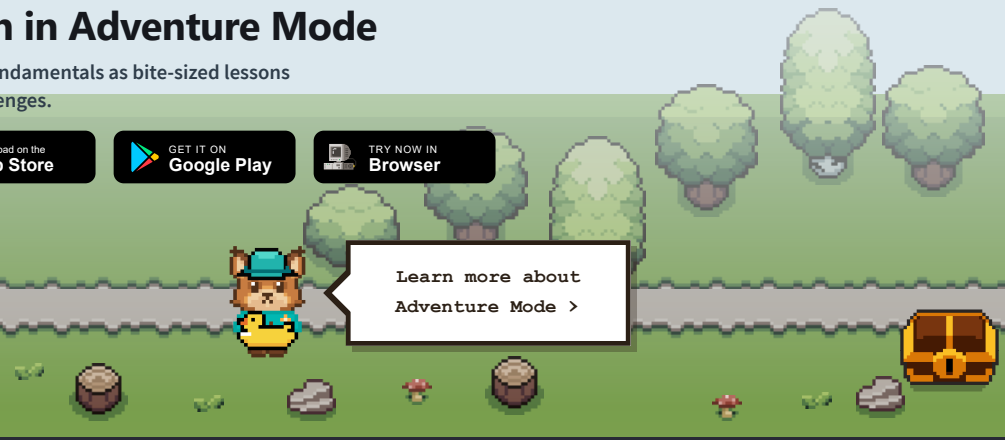


REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



w3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String fixed()

‹ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.fixed();
```

Try it Yourself »

Description

String fixed() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `fixed()` method returns a string embedded in a <tt> tag:

<tt>string</tt>

The <tt> tag is not supported in HTML5.

Syntax

```
string.fixed()
```

Parameters

NONE

Return Value

A string embedded in a <tt> tag.

‹ JavaScript String Reference



REMOVE ADS



FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String fontcolor()

‹ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.fontcolor("green");
```

Try it Yourself »

Description

String fontcolor() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `fontcolor()` method returns a string embedded in a tag:

string

The <font> tag is not supported in HTML5.

Syntax

```
string.fontcolor("color")
```

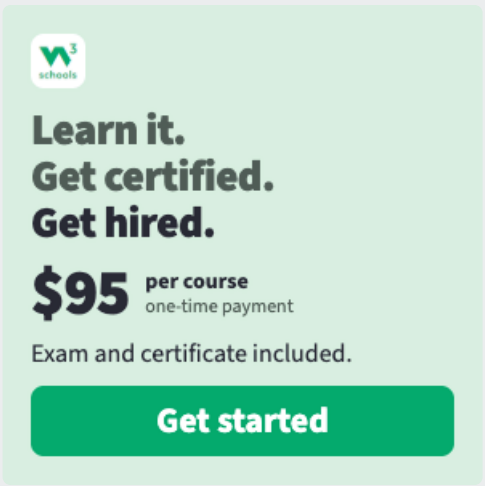
Parameters

Parameter	Description
color	Required. A color value in rgb or hex.

Return Value

A string embedded in the <font> tag.

❮ JavaScript String Reference



REMOVE ADS

Python - Get Started - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

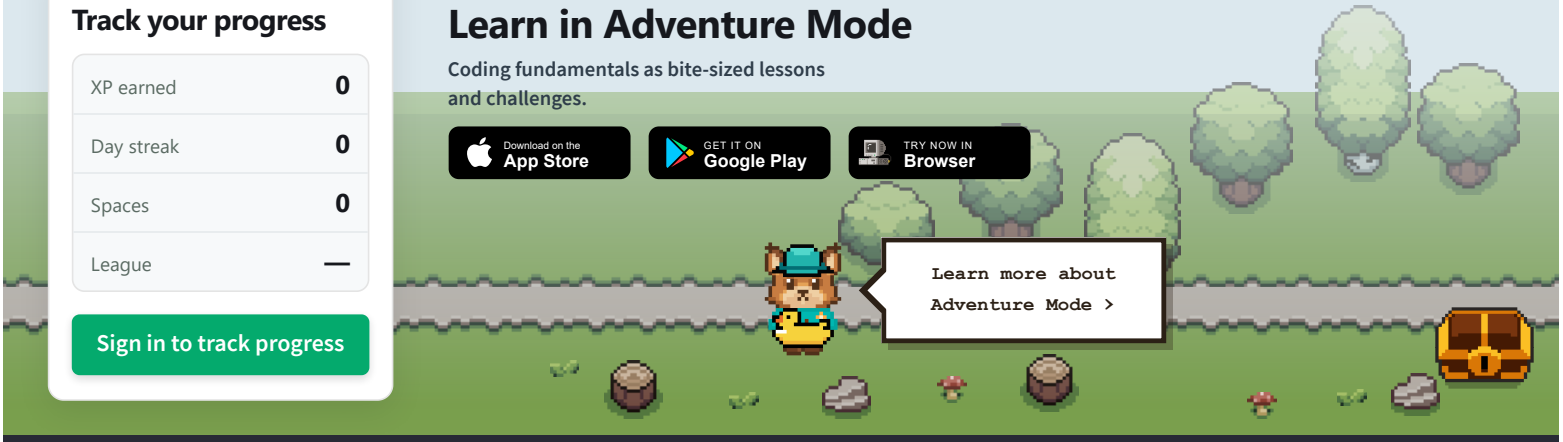
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String fontsize()

‹ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.fontsize(6);
```

Try it Yourself »

Description

String fontsize() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `fontsize()` method returns a string embedded in a tag:

string

The <font> tag is not supported in HTML5.

Syntax

```
string.fontsize(size)
```

Parameters

Parameter	Description
size	Required. Font size. An integer between 1 and 7.

Return Value

A string embedded in the <font> tag.

‹ JavaScript String Reference



REMOVE ADS

Python - Syntax - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

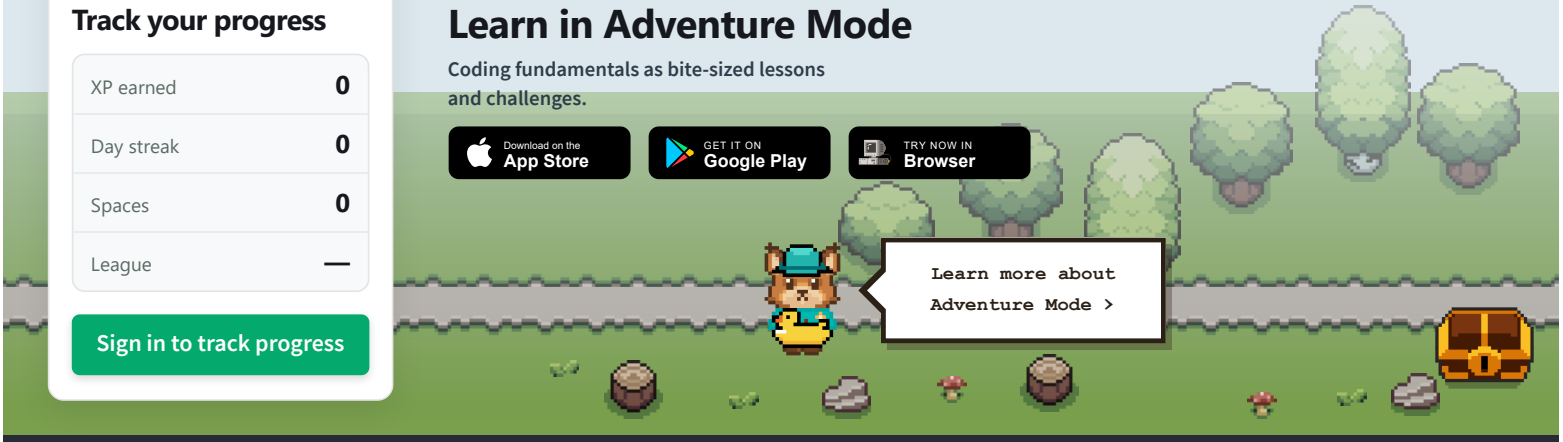
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String italics()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.italics();
```

Try it Yourself »

Description

String italics() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `italics()` method returns a string embedded in an tag:

string

Syntax

```
string.italics()
```

Parameters

NONE

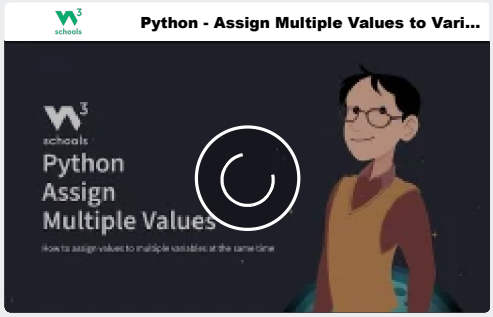
Retun Values

A string embedded in an <i> tag.

‹ JavaScript String Reference



REMOVE ADS



FREE ACCOUNT
Track your progress

NEW
Learn in Adventure Mode

XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference

HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String link()

‹ JavaScript String Reference

Example

```
let text = "Free Web Building Tutorials!";
let result = text.link("https://www.w3schools.com");
```

Try it Yourself »

Description

String link() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `link()` method returns a string embedded in an <a> tag:

string

Syntax

```
string.link("url")
```

Parameters

Parameter	Description
url	Required. The URL to link to.

Return Value

A string embedded in an <a> tag.

‹ JavaScript String Reference



REMOVE ADS



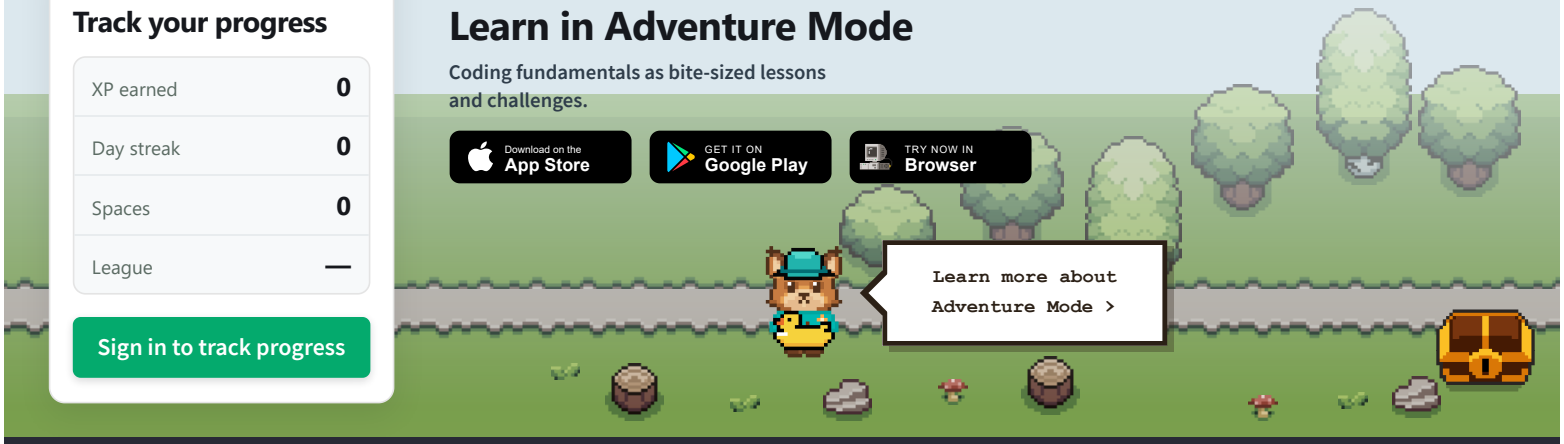
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String small()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.small();
```

Try it Yourself »

Description

String small() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `small()` method returns a string embedded in a <small> tag:

<small>string</small>

Syntax

```
string.small()
```

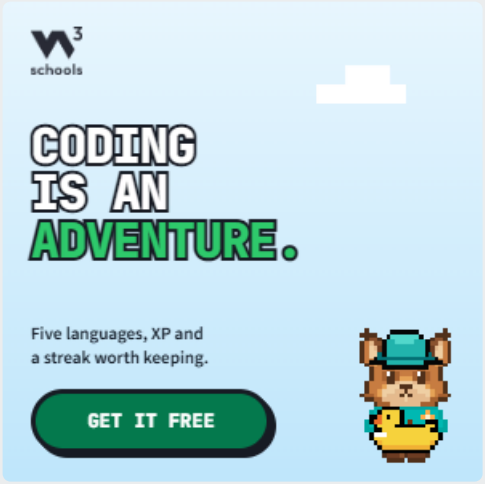
Parameters

NONE

Return Value

A string embedded in a <small> tag.

‹ JavaScript String Reference



REMOVE ADS



FREE ACCOUNT
Track your progress

NEW
Learn in Adventure Mode

XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference

HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript while Loop

‹ Previous JavaScript Statements Next ›

Example

Loop a code block as long as a i is less than 5:

```
let text = "";
let i = 0;
while (i < 5) {
  text += i + "<br>";
  i++;
}
```

Try it Yourself »

Loop (iterate over) an array to collect car names:

```
const cars = ["BMW", "Volvo", "Saab", "Ford"];
let text = "";
let i = 0;
while (i < cars.length) {
  text += cars[i] + "<br>";
  i++;
}
```

Try it Yourself »

The loop starts in position 0 (`let i = 0`).

The loop increments `i` for each run (`i++`).

The loop runs as long as `i < cars.length`.

More examples below.

Description

SHARE

The `while` statement creates a loop (araund a code block) that is executed while a condition is `true`.

The loop runs while the condition is `true`. Otherwise it stops.

See Also:

The JavaScript while Tutorial

JavaScript Loop Statements

Statement	Description
break	Breaks out of a loop
continue	Skips a value in a loop
while	Loops a code block while a condition is true
do...while	Loops a code block once, and then while a condition is true
for	Loops a code block while a condition is true
for...of	Loops the values of any iterable
for...in	Loops the properties of an object

Syntax

```
while (condition) {
  code block to be executed
}
```

Parameters

Parameter	Description

condition	Required. The condition for running the code block. If it returns true, the code clock will start over again, otherwise it ends.

Note

If the condition is always true, the loop will never end. This will crash your browser.

If you use a variable in the condition, you must initialize it before the loop, and increment it within the loop. Otherwise the loop will never end. This will also crash your browser.

REMOVE ADS

More Examples

Loop over an array in descending order (negative increment):

```
const cars = ["BMW", "Volvo", "Saab", "Ford"];
let text = "";
let len = cars.length;
while (len--) {
  text += cars[len] + "<br>";
}
```

Try it Yourself »

Using break - Loop through a block of code, but exit the loop when i == 3:

```
let text = "";
let i = 0;
while (i < 5) {
  text += i + "<br>";
  i++;
  if (i == 3) break;
}
```

Try it Yourself »

Using continue - Loop through a block of code, but skip the value 3:

```
let text = "";
let i = 0;
while (i < 5) {
  i++;
  if (i == 3) continue;
  text += i + "<br>";
}
```

Try it Yourself »

Browser Support

`while` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

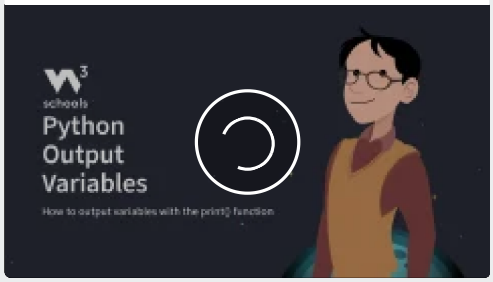
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Statements Next ›

W3schools
JS
Get Certified in JavaScript
Boost your career with an official certification
$95
Get certified

REMOVE ADS

Python - Output Variables with print() - ...



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

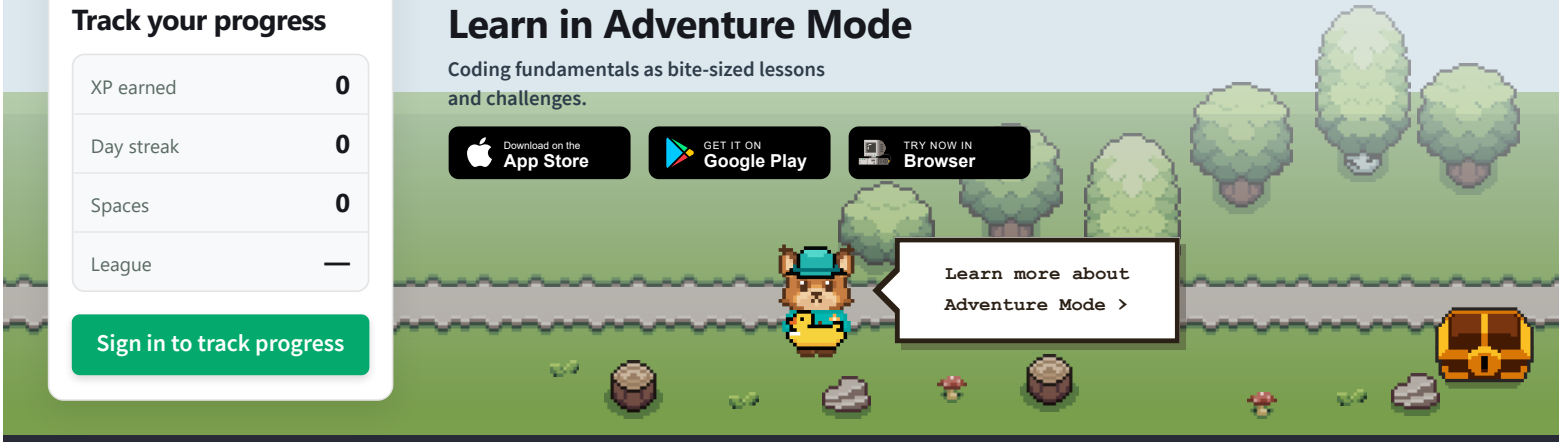
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String at()

‹ Previous JavaScript String Reference Next ›

Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at(0);
```

Try it Yourself »

Get the first character of the text:

```
let text = "W3Schools";
let character = text[0];
```

Try it Yourself »

More examples below.

Description

SHARE

The `at()` method returns an indexed character from a string.

The `at()` method returns the same as `[]`.

Syntax

```
string.at(index)
```

Parameters

Parameter	Description
index	Optional. The index (position) of the character to be returned. Default is 0. -1 returns the last character.

Return Value

Type	Description
A String	A single character. The character in the given position (index) in the string.

Related Pages

JavaScript Strings

JavaScript String Methods

JavaScript String Search

Browser Support

JavaScript String `at()` is supported in all browsers since March 2022:

				
Chrome 66	Edge 79	Firefox 61	Safari 12	Opera 50
Jul 2021	Jul 2021	Jul 2021	Mar 2022	Aug 2021

REMOVE ADS

More Examples

Get the first character of the text:

```
let text = "W3Schools";
let character = text.at();
```

Try it Yourself »

Get the last character of the text:

```
let text = "W3Schools";
let character = text.at(-1);
```

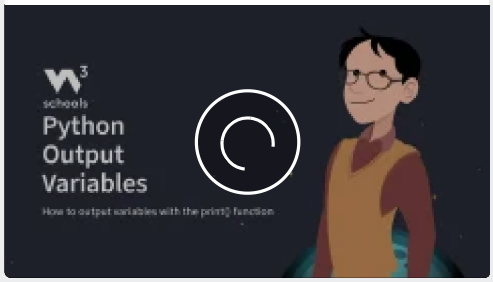
Try it Yourself »

‹ Previous JavaScript String Reference Next ›

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS

Python - Output Variables with print() - ...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

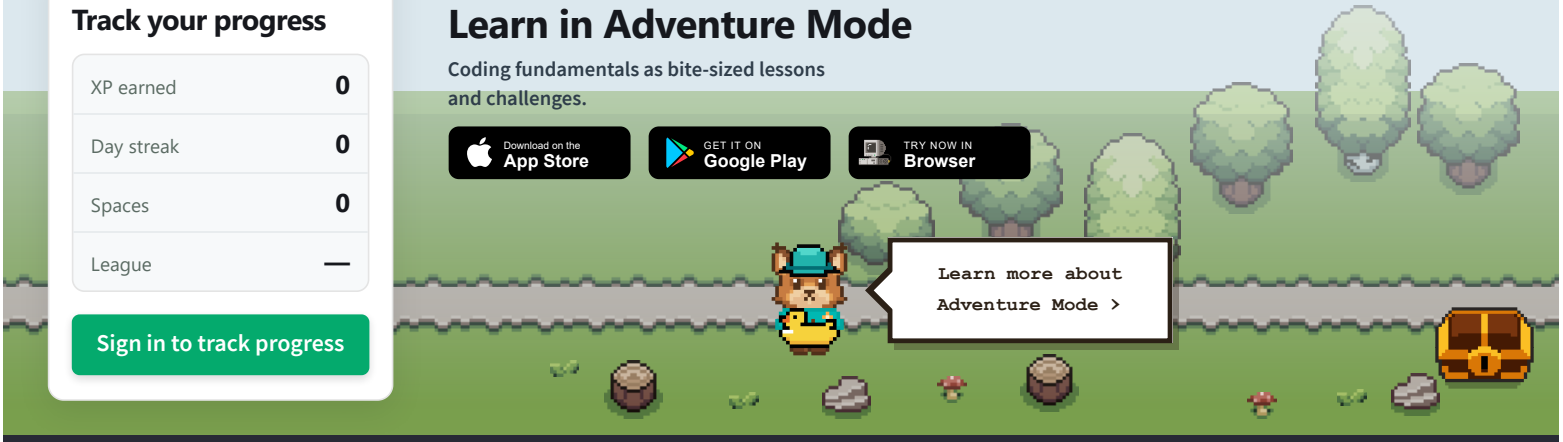
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String strike()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.strike();
```

Try it Yourself »

Description

String strike() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `strike()` method returns a string embedded in a <strike> tag:

<strike>string</strike>

The <strike> tag is not supported in HTML5.

Syntax

```
string.strike()
```

Parameters

NONE

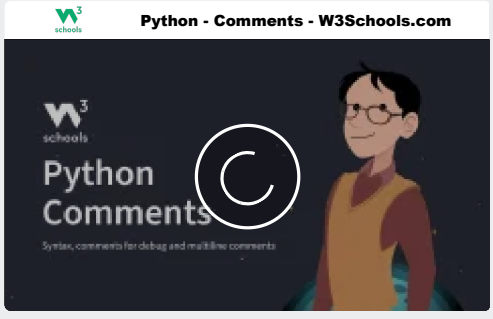
Return Value

A string embedded in a <strike> tag.

‹ JavaScript String Reference



REMOVE ADS



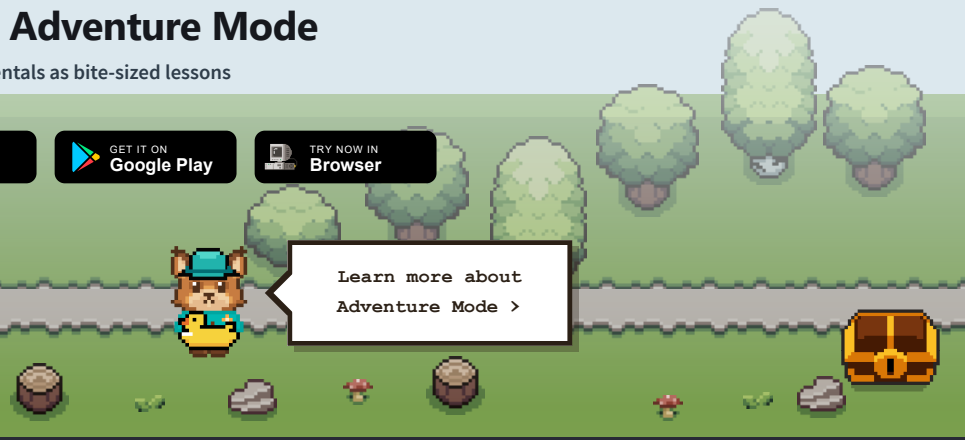
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String sub()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.sub();
```

Try it Yourself »

Description

String sub() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `sub()` method returns a string embedded in a <sub> tag:

_{string}

Syntax

```
string.sub()
```

Parameters

NONE

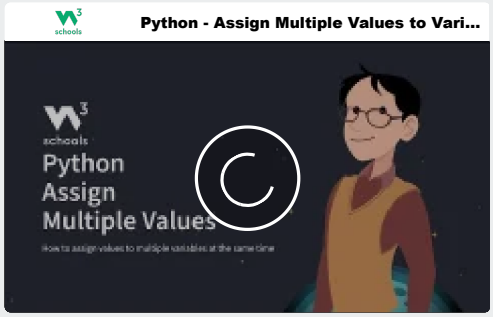
Return Value

A string embedded in a <sub> tag.

‹ JavaScript String Reference

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS

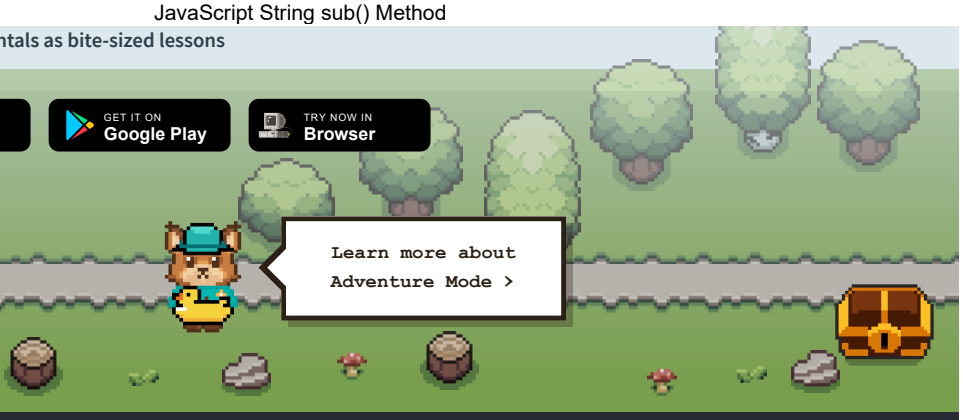


FREE ACCOUNT
Track your progress

NEW
Learn in Adventure Mode

XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference

HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript String sup()

❮ JavaScript String Reference

Example

```
let text = "Hello World!";
let result = text.sup();
```

Try it Yourself »

Description

String sup() is deprecated in JavaScript.

Avoid using it.

It may cease to work in your browser at any time.

The `sup()` method returns a string embedded in a <sup> tag:

^{string}

Syntax

```
string.sup()
```

Parameters

NONE

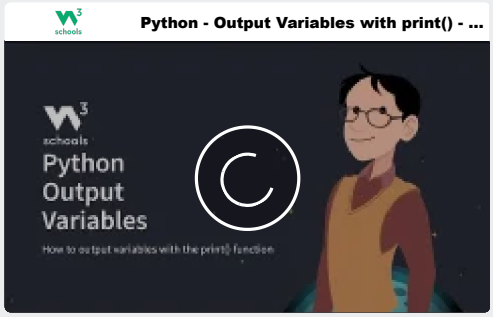
Return Value

A string embedded in a <sup> tag

‹ JavaScript String Reference



REMOVE ADS



FREE ACCOUNT
Track your progress

NEW
Learn in Adventure Mode

XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference

HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

Complete JavaScript References

‹ Home

Next ›

Revised June 2026

JavaScript Primitives

SHARE

Primitives | String | Number | Boolean | BigInt | Symbol | undefined | null

Object Methods and Properties

Array | Boolean | Classes | Date | Error | Function | Global | Iterator | JSON | Map | Math | Number | Object | Operators | Precedence | Promise | Proxy | Reflect | RegExp | Set | Statements | String | Temporal | Typed Array

Window Objects with Examples

Window | History | Location | Navigator | Screen

HTML DOM Objects with Examples

Document | Element | Attributes | Events | Event Objects | Collection | Style

REMOVE ADS

Web APIs with Examples

Canvas | Console | Fetch | Fullscreen | Geolocation | History | MediaQuery

Storage | Validation

HTML Elements with Examples

a | abbr | address | area | article | aside | audio | b | base | bdo | blockquote | body | br | button | canvas | caption

cite | code | col | colgroup | datalist | dd | del | details | dfn | dialog | div | dl | dt | em | embed | fieldset

figcaption | figure | footer | form | head | header | h1 - h6 | hr | html | i | iframe | img | ins | input button

input checkbox | input color | input date | input datetime | input datetime-local | input email | input file | input hidden

input image | input month | input number | input password | input radio | input range | input reset | input search | input submit

input text | input time | input url | input week | kbd | label | legend | li | link | map | mark | menu | menuitem | meta

meter | nav | object | ol | optgroup | option | output | p | param | pre | progress | q | s | samp | script | section

select | small | source | span | strong | style | sub | summary | sup | table | tbody | td | tfoot | th | thead | tr

textarea | time | title | track | u | ul | var | video

Other

CSSStyleDeclaration | JS Conversion

JavaScript Tutorial

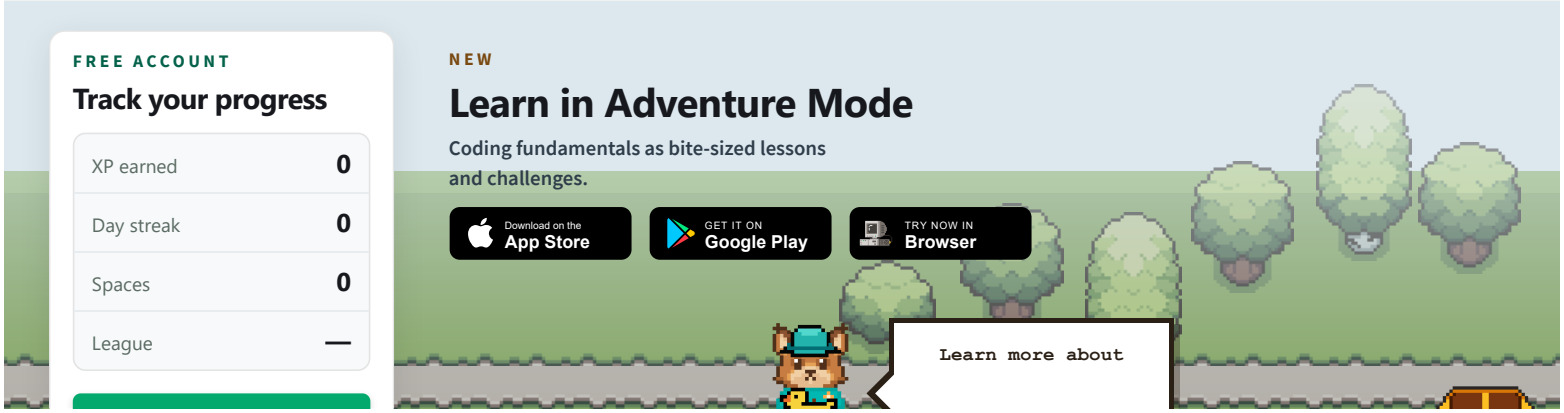
Visit Our JavaScript Tutorial »

‹ Home | Sign in to track progress | Next ›



REMOVE ADS







PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.